

Non-Standard Numbers as Global Sections of a Dedekind Reals Object

Thomas Booker
40347680

October 27, 2007

Abstract

We review the basic category theory and theorems up to the definition of an *elementary topos*, and we provide an example. Some topos theoretic work is involved in order to make explicit the construction of the Dedekind reals from a natural numbers object. We then prove that the *filterpower construction* on a topos $\mathcal{E}$, leads to a topos again. Finally, we exhibit the specific example of this that leads to the *non-standard reals* of Abraham Robinson. The global sections functor leads naturally to the ultrapower construction introduced first by Jerzy Łoś.

Contents

1	Introduction	2
2	The Basics	5
2.1	Useful Theory	5
2.2	Toposes	10
2.2.1	A Simple Example	13
3	The Beginnings of Real Analysis in a Topos	15
3.1	The Natural Numbers	15
3.2	Ring Objects in a Topos	20
3.3	The Integers	21
3.4	The Rationals	22
3.5	The Reals by Dedekind Cuts	24
3.5.1	A Logic Analogy	24
3.5.2	The Categorical Approach	27
3.5.3	The Dedekind Reals Object	30
4	The Filterpower Construction	31
4.1	Filters in Topology	31
4.1.1	Filters on $Sub(\mathbf{1})$	31
4.2	The Filterpower Construction	32
5	Non-Standard Real Numbers	38
5.1	Review of Basic Analysis	38
5.2	Global Sections	39
5.3	Topos of Sequences of Sets	39
5.4	Non-Standard Analysis	42
5.4.1	Requisite Theory	42
5.4.2	The Non-Standard Intermediate Value Theorem	44

Chapter 1

Introduction

Axiomatic set theory¹ is widely accepted as the best candidate we have as a foundation for all mathematics. From the concept of a set and membership we can begin to build up all mathematics as it is known today. One could construct elementary topology and from there all the traditional elements of analysis. On the other hand one could take an interest in sets with ‘structure’ leading to the concept of a monoid and all abstract algebra beyond. But what if you are not content with this view and would like your own model of set theory in which, say, the law of excluded middle did not hold? Enter topos theory. Building on the fundamental abstraction of mathematics known as category theory, topos theory seeks to provide a basis for the understanding of categories that exhibit certain properties of the category of sets² but can possess vastly different internal structure and logic.

Topos theory is one of those areas of mathematics that developed in two different ways due to a bifurcation in the creators respective schools of thought. This essentially comes down to the difference in origins of the definition of what is now known as a Grothendieck topos, and the concept of an elementary topos due to Lawvere and Tierney.

Towards the latter half of the 1950’s a large quantity of research was being pursued in algebraic geometry primarily due to the Weil conjectures. The problem was that André Weil had conjectured the existence of a particular theory of cohomology³, but was unable to provide a construction. Sometime around 1962, Alexander Grothendieck and Jean-Louis Verdier gave their definition of a topos which has come to be known as a Grothendieck topos. There are a couple of equivalent definitions but it essentially amounts to a category of sheaves on a site. A site is a category with a formal structure of “open covering”; the morphisms of the category are more versatile than mere inclusions between open subsets of a topological space. This then allowed Alexander Grothendieck to define something called the étale cohomology of a scheme which led to providing the construction that André Weil sought.

¹that is Zermelo-Fraenkel-(Choice) or ZFC set theory

²which admits ZFC set theory

³given the original name ‘Weil Cohomology’

The definition of what is now known as⁴ an elementary topos was delivered by Lawvere and Tierney in Zurich during August 1970. It began roughly sometime in 1963 when William Lawvere was searching for a new foundational base for mathematics using category theory. He started with the category of sets and separated out the pieces that made it ‘special’ from a categorical perspective. Perhaps the most radical difference was that, from the point of view of category theory, the membership of elements was not taken as primitive. What Lawvere discovered was that it is not the membership of an element to a set that makes the category of sets interesting; but more the ‘classification’ of the functions between them. The William Lawvere and Myles Tierney definition of an elementary topos revolves around the existence of what is known as a subobject classifier. To see why this is so critical to making the category of sets what it is, one need only consider the subobject classifier of this topos, namely, $\{0, 1\}$. This essentially allows one to say that the subsets of a set X are the same as functions defined on X to $\{0, 1\}$ (i.e. they are characteristic functions). The rest of the Lawvere and Tierney definition of an elementary topos is simply some demands on the existence of certain limits and “function objects”.

To see why the Lawvere and Tierney definition is so powerful, and to obtain some elaboration on the first paragraph, one should notice that it is the subobject classifier of the category of sets that gives it an underlying boolean logic structure. What is very nice is that the subobject classifier of an elementary topos need not have the structure of a boolean algebra. In fact in an arbitrary elementary topos the subobject classifier has a Heyting algebra structure which generally leads to, among many things, the law of excluded middle not holding.

In what seems completely unrelated to the above, we may cast our minds to the concept of an infinitesimal. These ‘numbers’ were (and are still) used heavily by physicists in order to talk about being infinitely close to something. This is often expressed as $x + dx$ and it is this dx that is referred to as an infinitesimal. For a long time their use was a topic of some controversy among mathematicians due to there being no consistent and rigorous reformulation of analysis using these ‘non-standard’ reals⁵. The first person to do this, and to whom it is now attributed, was Abraham Robinson in his book *Non-standard Analysis* which was first published in the early 1960’s. Robinson’s construction was heavily founded in terse mathematical logic, whereas we shall use the construction due to Jerzy Łoś (which relies on the existence of a non-principal ultrafilter ∇ of the natural numbers). The construction says that the non-standard reals are the ultrapower $\mathbb{R}^{\mathbb{N}}/\nabla$. That is, sequences (or vectors indexed by $\mathbb{N}$) $(r_1, r_2, r_3, \dots)$ modulo an equivalence relation relative to the elements of the ultrafilter ∇ . It is in this way that we identify the ‘standard reals’ with constant sequences such that we have the ‘standard reals’ sitting inside of the ‘non-standard reals’. Thus, a Robinson non-standard real is identified with a sequence that is different almost everywhere. Traditionally written ${}^*\mathbf{R}$, the non-standard reals form an ordered field where the partial ordering and field operations are performed modulo the ultrafilter ∇ .

⁴and has become what one might consider *the* definition of a topos today

⁵and thus no consistent model of non-standard calculus

Returning to the world of topos theory one may ask, now that we have a theory for categories possessing key properties of the category of sets, can we exhibit an internal structure paralleling the natural numbers? The characterization of so called *Natural Numbers Objects*⁶ and, while the categorical axiomatization is due to Lawvere, a lot of the theory about them is attributable to Peter Freyd who first presented the material in a 1972 publication of the ‘Bulletin of the Australian Mathematical Society’. Published after a series of lectures at the University of New South Wales, the paper lays the ground work to then begin considering abstract integers, rationals and reals objects in an arbitrary topos. While the construction of the integers and the rationals objects follow an analogous path to what one would expect classically, the two possible⁷ constructions of the reals require significantly more work. This is especially the case if one does not wish to use the internal logic language or ‘Mitchell-Bénabou language’ (which uses the Heyting algebra structure of the subobject classifier to exhibit a first-order predicate calculus). Instead, the required relations (subobjects) are constructed by hand and allow one to see more of the internal structure of the categorical analogue.

For a classical example of a category one can consider the specific functor category $\mathcal{E} = \text{Func}[\mathbf{N}, \mathbf{Set}]$. An object (functor) of this category assigns to each natural number a set in some universe and as such can be viewed as a sequence of sets indexed over the naturals. The morphisms (natural transformations) are then simply point-wise maps between sequences. This category is actually an example of a topos. In the category of sets the natural numbers, integers, rationals and reals objects are simply the traditional sets as objects themselves. Most importantly we have that the reals object $\mathbf{R}$ in the topos $\mathcal{E}$ is simply the constant sequence at the set of reals $\mathbb{R}$ in the category of sets. By Zorn’s lemma we can easily exhibit the existence of a non-principal ultrafilter ∇ on the natural numbers. We then introduce the filterpower of $\mathcal{E}$ modulo the ultrafilter ∇ in which the objects are those of $\mathcal{E}$ but the morphisms are modulo an equivalence relation defined using the subsets in ∇ . The projection functor taking us from $\mathcal{E}$ to the filterpower is said to be ‘logical’ in that it preserves all the topos structure of $\mathcal{E}$.

The global sections of an object in a topos is simply the hom-set of morphisms defined on the terminal object with codomain the object in question. Thus we take the global sections of the reals object in the filterpower of the afore described topos $\mathcal{E}$. This set is then said to be the *non-standard reals object* and directly parallels the construction due to Jerzy Łoś. This then allows one to reformulate traditional real and function analysis into their respective non-standard models yielding the consistent theory that Abraham Robinson once sought.

⁶NNO for short

⁷equivalence classes of Cauchy sequences and Dedekind cuts of the rationals

Chapter 2

The Basics

We shall begin by introducing (or perhaps reintroducing) the basic definitions and theorems required for our foray into Category Theory, finishing up with the definition of a Topos and a simple example.

2.1 Useful Theory

Definition A *category* $\mathcal{C}$ is a collection of objects $A, B, C, \dots \in ob \mathcal{C}$, together with a set of arrows as morphisms between said objects, $f : A \rightarrow B, \dots \in ar \mathcal{C}$. Additionally, we have that every object X has a corresponding identity arrow id_X as an endomorphism on itself. Finally there is an associative composition law ‘ $\circ$ ’ for arrows for which the id_X act as identities.

Some examples of common categories are **Set**, **Grp**, **Top**. Almost self explanatory, these are the category of sets as objects and functions as morphisms, groups and group homomorphisms, and topological spaces with the continuous functions between them. However, now that we have a category, the next natural thing to ask is if we took a category that had other categories as its objects; what would the morphisms be?

Definition A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ is a morphism of categories taking objects to objects and arrows to arrows. Functors must preserve identities and composition; that is,

$$\begin{aligned} F(id_A) &= id_{FA} \\ F(f \circ g) &= Ff \circ Fg \end{aligned}$$

Definition A *subfunctor* $G : \mathcal{C} \rightarrow \mathbf{Set}$ of $F : \mathcal{C} \rightarrow \mathbf{Set}$ is a functor such that,

1. For all $x \in \text{ob } \mathcal{C}$ we have $G(x) \subseteq F(x)$, and,
2. for all arrows $f : x' \rightarrow x$ we have that Gf is the restriction of Ff to $G(x')$.

Now that we have the concept of a morphism between categories, it makes sense to be able to move between functors as if they were objects. In category theory this is done in a special way so as to preserve certain properties.

Definition A *natural transformation* α between two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$ is a morphism of functors so that for each $A \in \text{ob } \mathcal{C}$ there is an assigned arrow $\alpha A : FA \rightarrow GA$ such that given a $g : A \rightarrow A'$ the diagram

$$\begin{array}{ccc} FA & \xrightarrow{\alpha A} & GA \\ Fg \downarrow & & \downarrow Gg \\ FA' & \xrightarrow{\alpha A'} & GA' \end{array}$$

commutes.

As an aside, we can now talk of the category $\mathbf{Func}[\mathcal{C}, \mathcal{D}]$, the functor category. This is where the objects are functors from $\mathcal{C}$ to $\mathcal{D}$ and the arrows are the natural transformations between them. As can be seen at the end of this chapter this is an example of a topos provided $\mathcal{D}$ is the category $\mathbf{Set}$ of sets.

In a category $\mathcal{C}$ an *element* of an object A defined on X is an arrow $f : X \rightarrow A$ and is often written $f \in^X A$. A *monomorphism* $f : X \rightarrow A$ is an element of A that is injective on elements of X .

If $u : X \rightarrow A$ is a monomorphism and $v : Y \rightarrow A$ then v is said to *factor* through u if there exists an arrow $k : X \rightarrow Y$ such that $u \circ k = v$.

Definition A *subobject* of A is an equivalence class of monomorphisms into A . Two monomorphisms $p : X \rightarrow A$ and $q : Y \rightarrow A$ are said to be equivalent if there exists an invertible morphism (isomorphism) $\kappa : X \rightarrow Y$ such that

$$\begin{array}{ccc} X & & \\ \downarrow \kappa & \searrow p & \\ & & A \\ & \nearrow q & \\ Y & & \end{array}$$

commutes.

It makes sense now to define the *subobject functor*, written as $Sub : \mathcal{C}^{op} \rightarrow \mathbf{Set}$, which takes an object in $\mathcal{C}$ and sends it to the set of all its subobjects. This functor becomes key in the definition of a topos as will be shown.

Lemma 2.1.1 (Yoneda's Lemma) *Let $\mathcal{C}$ be a locally small category (its Hom-sets are actual sets) and $F : \mathcal{C} \rightarrow \mathbf{Set}$ a functor. Yoneda's lemma says,*

$$\phi : \text{Nat}(\text{Hom}_{\mathcal{C}}(A, -), F) \rightarrow F(A)^1$$

defined by $\phi(\lambda) = (\lambda A)(id_A)$ is a natural isomorphism. Put another way, the number of ways to naturally transform into F is in one to one correspondence with the elements of $F(A)$.

Proof Let $f : A \rightarrow X$ be an arrow in $\mathcal{C}$. Given $u \in FA$, if we want to find its inverse image under ϕ , we are forced to define λ by $\lambda X \circ f = Ff \circ u$ in order for the standard natural transformation diagram

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(A, A) & \xrightarrow{\lambda A} & FA \\ \text{Hom}_{\mathcal{C}}(A, f) \downarrow & & \downarrow Ff \\ \text{Hom}_{\mathcal{C}}(A, X) & \xrightarrow{\lambda X} & FX \end{array}$$

to commute. It is easily seen that defining λ in this way makes it natural. This proves that ϕ is bijective. It is clearly natural in both A and F . ■

Lastly before we can define a product and categorical limit we need the concept of a universal element for a functor. If $F : \mathcal{C} \rightarrow \mathbf{Set}$ is a functor then $u \in FA$ is said to be a *universal element* for F if and only if for all $B \in \text{ob } \mathcal{C}$ and all elements $j \in FB$, there exists a unique $g : A \rightarrow B$ such that $Fg \circ u = j$. A functor $F : \mathcal{C} \rightarrow \mathbf{Set}$ is said to be *representable* if it is naturally isomorphic to $\text{Hom}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}$. By Yoneda's lemma we see that F is representable precisely when the pair (A, u) as in the lemma is a universal element for F .

¹The Hom-set functor here takes every object $X \in \text{ob } \mathcal{C}$ to the set $\text{Hom}(A, X)$ of elements of X defined on A

Before we can talk of limits we need the concept of a *cone for a functor*. Given a functor $F : \mathcal{D} \rightarrow \mathcal{C}$, a cone (N, ψ) for F consists of an object N of $\mathcal{C}$ together with a set of morphisms $\psi_X : N \rightarrow F(X)$, for each $X \in \text{ob } \mathcal{D}$, such that for every $f : X \rightarrow Y$ in $\mathcal{D}$ we have $Ff \circ \psi_X = \psi_Y$. That is, ψ is a natural transformation from the constant functor at N to F .

Definition The *limit* of a functor $F : \mathcal{D} \rightarrow \mathcal{C}$ is a universal cone (L, ψ) . “Universal” means for any other cone (N, ϕ) there exists unique morphism $u : N \rightarrow L$ making the diagram

$$\begin{array}{ccccc}
 & N & & & \\
 & \downarrow u & & & \\
 & L & & & \\
 \phi_X \swarrow & & \searrow \phi_Y & & \\
 F(X) & \xrightarrow{Ff} & F(Y) & &
 \end{array}$$

(Note: The diagram above is a simplified representation of the one in the image. The image shows a triangle with N at the top, L in the middle, and $F(X)$ and $F(Y)$ at the bottom. Arrows: $N \rightarrow L$ (dashed, labeled u), $N \rightarrow F(X)$ (solid, labeled ϕ_X), $N \rightarrow F(Y)$ (solid, labeled ϕ_Y), $L \rightarrow F(X)$ (solid, labeled ψ_X), $L \rightarrow F(Y)$ (solid, labeled ψ_Y), $F(X) \rightarrow F(Y)$ (solid, labeled Ff).

commute. That is, each ϕ_X of any other commutative cone factors over L with the unique factorisation u .

A colimit is just as it sounds, a universal co-cone for the functor F where the prefix ‘co-’ means what it usually does in category theory; that is, “Reverse all arrows”. Put explicitly it is (L, ψ) such that for any other co-cone (N, ϕ) there exists unique morphism $u : L \rightarrow N$ making the diagram

$$\begin{array}{ccccc}
 F(X) & \xrightarrow{Ff} & F(Y) & & \\
 \searrow \psi_X & & \swarrow \psi_Y & & \\
 & L & & & \\
 \swarrow \phi_X & & \searrow \phi_Y & & \\
 & N & & &
 \end{array}$$

(Note: The diagram above is a simplified representation of the one in the image. The image shows a triangle with $F(X)$ and $F(Y)$ at the top, L in the middle, and N at the bottom. Arrows: $F(X) \rightarrow F(Y)$ (solid, labeled Ff), $F(X) \rightarrow L$ (solid, labeled ψ_X), $F(Y) \rightarrow L$ (solid, labeled ψ_Y), $L \rightarrow N$ (dashed, labeled u), $L \rightarrow F(X)$ (solid, labeled ϕ_X), $L \rightarrow F(Y)$ (solid, labeled ϕ_Y).

commute.

We will now quickly look at three specific cases of limit known as product, pullback and equalizer. These are the three kinds of limits that appear when we meet the idea of a topos.

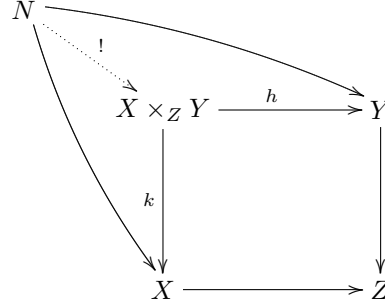
Definition Let $A, B \in \text{ob } \mathcal{C}$. The *product* of A and B in the category $\mathcal{C}$ is an object $A \times B$ together with two projections $\pi_1 : A \times B \rightarrow A$ and $\pi_2 : A \times B \rightarrow B$ such that, for any $u \in {}^T A$ and $v \in {}^T B$, there exists a unique element $(u, v) \in {}^T A \times B$ such that the diagram

$$\begin{array}{ccccc}
 & T & & & \\
 u \swarrow & & \searrow v & & \\
 A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B
 \end{array}$$

(Note: The diagram above is a simplified representation of the one in the image. The image shows a triangle with T at the top, A and B at the bottom, and $A \times B$ in the middle. Arrows: $T \rightarrow A$ (solid, labeled u), $T \rightarrow B$ (solid, labeled v), $T \rightarrow A \times B$ (dashed, labeled (u, v)), $A \times B \rightarrow A$ (solid, labeled π_1), $A \times B \rightarrow B$ (solid, labeled π_2).

commutes.

Definition A *pullback* is a triple (P, h, k) forming the limit of a finite diagram of three objects with two arrows having a common codomain; that is, a universal commutative cone based at P , written here as $X \times_Z Y$, such that the following diagram



commutes. A pullback is sometimes referred to as a fibred product.

Definition An *equalizer* is the limit of a diagram of two objects and two morphisms between them sharing the same codomain. It is the universal cone (P, t) such that the following diagram

$$P \xrightarrow{t} X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} Y$$

commutes.

The duals of the above three constructions are obviously coproduct, pushout and coequalizer. The final ingredient we require in order to say what it is for a category to be finitely complete is the notion of a terminal object.

Definition The *terminal object* in a category $\mathcal{E}$ is an object $\mathbf{1}$ that is terminal for every $X \in \text{ob } \mathcal{E}$. That is, for each object X there is precisely one morphism $\rho_X : X \rightarrow \mathbf{1}$. The natural dual notion of this is the initial object which is said to be coterminal.

If we once again take **Set** as our category we see that a terminal object is trivially any singleton² set.

Definition A category $\mathcal{C}$ is said to be *finitely complete* when $\mathcal{C}$ has a terminal object together with all products and equalizers³.

Naturally one also has the dual notion of a finitely cocomplete category. The idea of completeness plays a pivotal role in the defining of a topos.

²a set with precisely one object is called a singleton.

³here “pullbacks” can be substituted for “products and equalizers”.

2.2 Toposes

In this section we develop the remaining theory that is central to the defining of a topos. The following theory comprises fundamental properties of the category **Set** of sets.

Definition For a fixed $A \in \text{ob } \mathcal{E}$, let $- \times A : \mathcal{E} \rightarrow \mathcal{E}$ be the endofunctor such that given an $f : B \rightarrow B'$ we have an arrow $f \times A$ defined as $(f \circ \pi_1, \pi_2) : B \times A \rightarrow B' \times A$. The *power object* $\wp A$ of A is an object of $\mathcal{E}$ that makes $\text{Sub}(- \times A)$ representable. That is, there is a natural isomorphism

$$\text{Hom}_{\mathcal{E}}(-, \wp A) \cong \text{Sub}(- \times A)$$

so that the following diagram

$$\begin{array}{ccc} \text{Hom}_{\mathcal{E}}(B, \wp A) & \xrightarrow{\beta(A, B)} & \text{Sub}(B \times A) \\ \text{Hom}_{\mathcal{E}}(f, \wp A) \downarrow & & \downarrow \text{Sub}(f \times A) \\ \text{Hom}_{\mathcal{E}}(B', \wp A) & \xrightarrow{\beta(A, B')} & \text{Sub}(B' \times A) \end{array}$$

commutes.

In order to have a better picture of the mechanics behind this, consider the following diagram

The diagram illustrates the relationship between objects B , B' , A , their products with A , and the power object $\wp A$. It shows a 3x3 grid of objects. The top row consists of $\wp A$, $B \times A$, and A . The middle row consists of B , $B \times A$, and A . The bottom row consists of $\wp A$, $B' \times A$, and A . Arrows include: $B \times A \rightarrow B$ (π_1), $B \times A \rightarrow A$ (π_2), $B \rightarrow B'$ (f), $B' \times A \rightarrow B'$ (π_1'), $B' \times A \rightarrow A$ (π_2), $B \rightarrow \wp A$ (j'), $B' \rightarrow \wp A$ (q'), $B \times A \rightarrow B' \times A$ ($f \times A$), and various inclusion/exclusion arrows like $i, j, p, q, i', j', p', q'$.

in which we see that the subobjects of the product are in one to one correspondence with morphisms onto the power object.

In the category **Set**, if given a set A , we can take the power set $\wp A$. We then have that elements defined on the terminal object are the inclusions into a subset of A and subobjects of A consist of one inclusion into a subset of A and injective maps for which their image is that subset. The isomorphism follows.

Before we talk of cartesian closed categories, we need to know what an exponential object is.

Definition Given $Z, Y \in \text{ob } \mathcal{E}$, their *exponential object* Z^Y is an object of $\mathcal{E}$ together with an evaluation morphism $eval : Z^Y \times Y \rightarrow Z$ such that there exists a unique morphism $\hat{f} : X \rightarrow Z^Y$ making the diagram

$$\begin{array}{ccc} Z^Y \times Y & \xrightarrow{eval} & Z \\ \hat{f} \times id_Y \uparrow & \nearrow f & \\ X \times Y & & \end{array}$$

commute. If every Z has an exponential object then the implicit endomorphic functor sending Z to Z^Y is right adjoint to the endomorphic functor $- \times Y$, thus

$$Hom_{\mathcal{E}}(X, Z^Y) \cong Hom_{\mathcal{E}}(X \times Y, Z).$$

Definition A category $\mathcal{E}$ is said to be *cartesian closed* if and only if it admits the following,

1. a terminal object;
2. for any two $X, Y \in \text{ob } \mathcal{E}$, a product $X \times Y$ in $\mathcal{E}$;
3. for any two $Z, Y \in \text{ob } \mathcal{E}$, an exponential Z^Y in $\mathcal{E}$.

A basic example of a cartesian closed category is the category **Set**. We can take finite products of sets which are back in **Set**. Moreover given any function $f : X \times Y \rightarrow Z$ we can naturally associate⁴ it to the curried function⁵ $\hat{f}(x)(y) : X \rightarrow Z^Y$. Here the *eval* map simply sends (g, y) to $g(y)$.

The final piece of the puzzle is something called the subobject classifier. It takes a main role in the second definition of a topos.

Definition Given a category $\mathcal{E}$, an object Ω is said to be a *subobject classifier* when it is equipped with a morphism $tr : \mathbf{1} \rightarrow \Omega$ such that, for all monomorphisms $m : S \rightarrowtail X$, there exists a unique morphism $\chi_m : X \rightarrow \Omega$ with the diagram

$$\begin{array}{ccc} S & \xrightarrow{m} & X \\ \downarrow & & \downarrow \exists! \chi_m \\ \mathbf{1} & \xrightarrow{tr} & \Omega \end{array}$$

a pullback. The abbreviation 'tr' stands for true. Clearly this determines an isomorphism

$$Hom_{\mathcal{E}}(X, \Omega) \cong Sub(X)$$

which says that the morphisms into Ω are in natural correspondence with the subobjects of X .

⁴In the sense of functions between sets.

⁵A function with some arguments fixed to obtain a function in terms of the other variables.

We now have all the pieces we require to define what we mean by a topos. On a fundamental level it is a category that exhibits certain properties of the category of sets, namely finite completeness, and that we can always form a powerset of a given set. A general topos is simply an abstraction of this.

Definition A category $\mathcal{E}$ is said to be a *topos* if it has finite limits and every object has a powerobject.

While this definition is all well and good it is often not easy to show the existence of power-objects for each object in a general category. We therefore consider the following theorem.

Theorem 2.2.1 *A category $\mathcal{E}$ is a topos if and only if:*

1. *It is finitely complete,*
2. *It is cartesian closed,*
3. *It has a subobject classifier.*

Proof We shall only prove “only if”. Immediately we have the implication of finite completeness. Now, taking Ω to be the subobject classifier, we see that since $\mathcal{E}$ is cartesian closed we can take the exponential object Ω^X for each $X \in \text{ob } \mathcal{E}$; then

$$\text{Hom}(B, \Omega^X) \cong \text{Hom}(B \times X, \Omega) \cong \text{Sub}(B \times X).$$

In other words every object X has a powerobject Ω^X . ■

The converse is somewhat more involved and requires some significant theoretic steps to be taken.

Lastly we define a special case of a functor between toposes.

Definition Let $\mathcal{E}$ and $\mathcal{E}'$ be toposes. A functor $\mathcal{L} : \mathcal{E} \rightarrow \mathcal{E}'$ is said to be *logical* if it preserves all finite limits, exponentials and the subobject classifier.

It is clear that logical functors preserve the underlying structure of a topos. We will meet such a functor in chapter 4.

2.2.1 A Simple Example

To illustrate the afore discussed theory we now turn to a simple example known as the *Functor Category*. The functor category $Func[\mathcal{C}^{op}, \mathbf{Set}]$ has as objects all the functors from $\mathcal{C}^{op6}$ onto $\mathbf{Set}$ and as arrows the natural transformations between the functors. We take for granted that $\mathbf{Set}$ is a topos.⁷

Theorem 2.2.2 $Func[\mathcal{C}^{op}, \mathbf{Set}]$ is a topos.

Proof The functor that sends every object of $\mathcal{C}$ to a singleton in $\mathbf{Set}$ and every morphism of $\mathcal{C}$ to the identity on that singleton is the terminal object in $Func[\mathcal{C}^{op}, \mathbf{Set}]$. Additionally all limits in $Func[\mathcal{C}^{op}, \mathbf{Set}]$ are calculated point-wise in $\mathbf{Set}$, which is complete (and so they exist), for each $C \in ob\mathcal{C}$. Hence we have that $Func[\mathcal{C}^{op}, \mathbf{Set}]$ is complete. All that remains to be seen now is the existence of exponentials and a subobject classifier.

For F^G to be an exponential object it must satisfy

$$Hom_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(H, F^G) \cong Hom_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(H \times G, F)$$

for every representable functor $H = Hom_{\mathcal{C}}(-, C)$. By Yoneda we are forced to define

$$F^G(C) = Hom_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(Hom_{\mathcal{C}}(-, C) \times G, F),$$

where the evaluation map $eval : F^G \times G \rightarrow F$ is defined by $eval_C(\phi, x) = \phi_C(id_C, x) \in F(C)$ for all $C \in ob\mathcal{C}$, $\phi : Hom_{\mathcal{C}}(-, C) \times G \rightarrow F$ and $x \in G(C)$. Then $eval$ is a natural transformation and for every f there is a unique $\hat{f}$ such that the diagram in the definition commutes.

All that is left to show is that there is a subobject classifier. For one to exist it must classify the subfunctors of a given functor F . It must therefore classify the subfunctors of F 's representation $Hom_{\mathcal{C}}(-, A)$. So we must have,

$$\begin{aligned} Sub_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(F) &\cong Sub_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(Hom_{\mathcal{C}}(-, A)) \\ &\cong Hom_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(Hom_{\mathcal{C}}(-, A), \Omega) \\ &= Nat(Hom_{\mathcal{C}}(-, A), \Omega) \end{aligned}$$

By Yoneda we are forced to define

$$\Omega(A) = Sub_{Func[\mathcal{C}^{op}, \mathbf{Set}]}(Hom_{\mathcal{C}}(-, A)),$$

the set of subfunctors of $Hom_{\mathcal{C}}(-, A)$ which are called A -sieves. More explicitly, an A -sieve S is a set of morphisms into A such that if $u : X \rightarrow A$ is in S and $g : Y \rightarrow X$ is any morphism, then $u \circ g$ is in S . Given a morphism $f : A' \rightarrow A$, the pullback of a sieve S along f is a new sieve f^*S of $Hom_{\mathcal{C}}(-, A')$. f^*S can also be viewed as the image of the fibre product $S \times_{Hom_{\mathcal{C}}(-, A)} Hom_{\mathcal{C}}(-, A')$ by the natural projection π in the following pullback diagram

$$\begin{array}{ccc} S \times_{Hom_{\mathcal{C}}(-, A)} Hom_{\mathcal{C}}(-, A') & \xrightarrow{\pi} & Hom_{\mathcal{C}}(-, A') \\ \downarrow & & \downarrow Hom_{\mathcal{C}}(A', f) \\ S & \longrightarrow & Hom_{\mathcal{C}}(-, A) \end{array}$$

⁶all arrows reversed.

⁷which has essentially been shown through the course of the afore theory.

That is $f^*S = \{g : f \circ g \in S\}$.

So we now have the only possible candidate for the subobject classifier of $Func[\mathcal{C}^{op}, \mathbf{Set}]$ being a functor Ω with the following properties,

1. for an $A \in ob\mathcal{C}$ have $\Omega(A) = \{S : S \text{ is a sieve on } A\}$
2. for a $f : A' \rightarrow A$ have $\Omega f = \Omega(A) \rightarrow \Omega(A')$ and $f^*S = \{g : f \circ g \in S\}$

For this to be the functor we are looking for, it must have an accompanying morphism tr making the square in the definition a pullback. If we take an $A \in ob\mathcal{C}$ then the set of all arrows onto A is called the maximal sieve on A . If we take the collection of all these for each $A \in ob\mathcal{C}$ and glue them together we yield a morphism $tr : \mathbf{1} \rightarrow \Omega$. To show that this is all we require, consider a subfunctor G of $F \in Func[\mathcal{C}^{op}, \mathbf{Set}]$. Given any morphism $f : A \rightarrow B \in \mathcal{C}$ we have a morphism $Ff : F(B) \rightarrow F(A)$ in $\mathbf{Set}$. If we take an $\eta \in F(B)$ we want to ensure that $Ff \circ \eta \in G(A) \subseteq F(A)$, so we define for each $\eta \in F(B)$,

$$\psi_B(\eta) = \{g : Fg \circ \eta \in G(dom\,g)\}.$$

We see that $\psi_B(\eta)$ is a sieve on B^8 and if it is taken to range over all objects in $\mathcal{C}$ then it forms a pointwise description of a natural transformation $\psi : F \rightarrow \Omega$. We see now that the subfunctor G is the pullback of ψ along tr making the diagram

$$\begin{array}{ccc} G & \longrightarrow & F \\ \downarrow & & \downarrow \psi \\ \mathbf{1} & \xrightarrow{tr} & \Omega \end{array}$$

commute. ■

Note that we can consider the functor category $Func[\mathcal{O}(X)^{op}, \mathbf{Set}]$ where $\mathcal{O}(X)^{op}$ is the opposite category of the topological space X ; that is, the category where the objects are open subsets of X and the arrows are inclusions. This functor category is known as $Psh(X)$, the category of presheaves on X , a very classic example of a topos.

⁸it is the maximal sieve if and only if $\eta \in G(B)$

Chapter 3

The Beginnings of Real Analysis in a Topos

In traditional set theory one of the first steps to be taken is the construction of the natural numbers; classically done via Peano's axioms. Once one has an idea of the natural numbers, it is then a small step to construct the integers (as equivalence classes of pairs of natural numbers) and from them the rationals (as equivalence classes of pairs of integers) before finally constructing the reals (as equivalence classes of Cauchy sequences of rational numbers or by Dedekind cuts). One can then begin to consider all the traditional elements of real analysis. The question is, how does one get to the analogue of the real numbers in an arbitrary topos.

3.1 The Natural Numbers

Definition A topos $\mathcal{E}$ is said to have a *natural numbers object* when there exists an object $\mathbf{N}$ together with arrows $0 : \mathbf{1} \rightarrow \mathbf{N}$ and $\text{succ} : \mathbf{N} \rightarrow \mathbf{N}$ such that for any other arrow $x_0 : \mathbf{1} \rightarrow A$ and $u : A \rightarrow A$ there exists a unique $s : \mathbf{N} \rightarrow A$ such that $s0 = x_0$ and $s \text{succ} = us$. That is, the following diagram

$$\begin{array}{ccccc} & & \mathbf{N} & \xrightarrow{\text{succ}} & \mathbf{N} \\ & \nearrow 0 & \vdots \exists! s & & \vdots \exists! s \\ \mathbf{1} & & & & \\ & \searrow x_0 & A & \xrightarrow{u} & A \end{array}$$

commutes. We often refer to the natural numbers object as an 'NNO'.

First we have some categorical properties of an NNO in an arbitrary topos $\mathcal{E}$ which are analogous to some of Peano's axioms.

Proposition 3.1.1 *If $\mathbf{1} \xrightarrow{0} \mathbf{N} \xrightarrow{suc} \mathbf{N}$ in a topos $\mathcal{E}$ is an NNO then the map $[0, suc] : \mathbf{1} + \mathbf{N} \rightarrow \mathbf{N}$ is an isomorphism.*

Proof Consider the coproduct $\mathbf{1} \xrightarrow{cop_1} \mathbf{1} + \mathbf{N} \xleftarrow{cop_2} \mathbf{N}$ where cop_1 and cop_2 are the coprojections. We then define the map $suc' : \mathbf{1} + \mathbf{N} \rightarrow \mathbf{1} + \mathbf{N}$ by the compositions $suc'.cop_1 = cop_2.0$ and $suc'.cop_2 = cop_2.suc$. Now, there exists a unique $s : \mathbf{N} \rightarrow \mathbf{1} + \mathbf{N}$ having the property that $s.0 = cop_1$ and $s.suc = suc'.s$. We now have (from the definition of NNO) the following diagram

$$\begin{array}{ccccc}
 & & \mathbf{N} & \xrightarrow{suc} & \mathbf{N} \\
 & \nearrow 0 & \vdots s & & \vdots s \\
 \mathbf{1} & \xrightarrow{cop_1} & \mathbf{1} + \mathbf{N} & \xrightarrow{suc'} & \mathbf{1} + \mathbf{N} \\
 & \searrow 0 & \downarrow [0, suc] & & \downarrow [0, suc] \\
 & & \mathbf{N} & \xrightarrow{suc} & \mathbf{N}
 \end{array}$$

We see that in the lower square

$$[0, suc].suc'.cop_1 = [0, suc].cop_2.0 = suc.0 = suc.[0, suc].cop_1$$

and,

$$[0, suc].suc'.cop_2 = [0, suc].cop_2.suc = suc.suc = suc.[0, suc].cop_2$$

yielding,

$$[0, suc].suc' = suc.[0, suc] \quad .$$

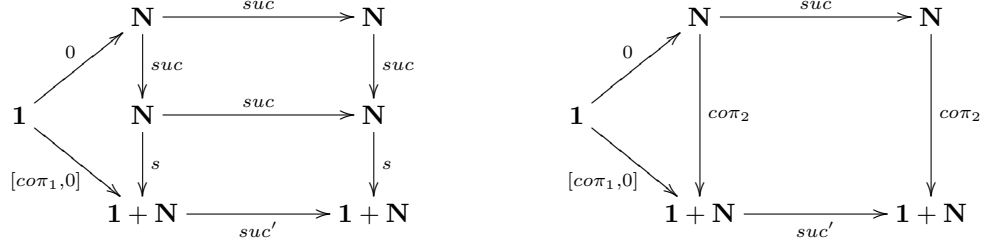
Clearly by the uniqueness of s above we have that $[0, suc].s = id_{\mathbf{N}}$. The other direction comes from

$$\begin{array}{ccccc}
 & & \mathbf{N} & \xrightarrow{suc} & \mathbf{N} \\
 & \nearrow 0 & \parallel & & \parallel \\
 \mathbf{1} & \xrightarrow{cop_2} & \mathbf{1} + \mathbf{N} & \xrightarrow{suc'} & \mathbf{1} + \mathbf{N} \\
 & \searrow cop_2.0 & \downarrow suc.s & & \downarrow suc.s \\
 & & \mathbf{1} + \mathbf{N} & \xrightarrow{suc'} & \mathbf{1} + \mathbf{N}
 \end{array}$$

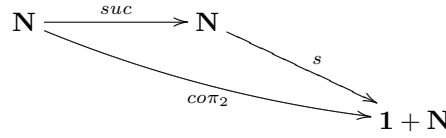
and applying uniqueness to obtain $suc.s = cop_2$ and so $[0, suc].s = id_{\mathbf{N}}$. ■

Theorem 3.1.2 *The map $\text{suc} : \mathbf{N} \rightarrow \mathbf{N}$ is mono.*

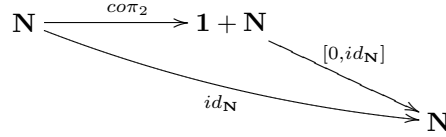
Proof We can form the following two commuting diagrams



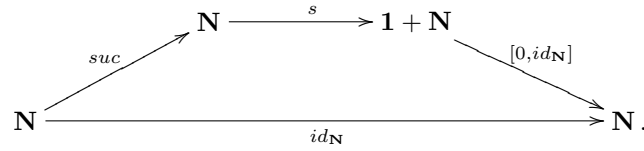
where suc' and s (the unique map in the definition) are defined as before. So clearly, from the right hand side of both diagrams, we have that the following diagram



commutes. However we can also form the following diagram

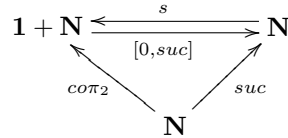


which clearly commutes. Bringing the two together yields



Hence suc is a coretraction. $\blacksquare$

Proof (Alternative)¹



where $[0, \text{suc}]$ has inverse s and $\text{co}\pi_2$ is mono forcing suc to be as well. $\blacksquare$

¹Relies on knowing that in a topos, coprojections are mono.

In order to construct the integers (additive inverses for elements of $\mathbf{N}$) and the rationals (multiplicative inverses for elements of $\mathbf{Z}$) we require addition, multiplication and exponentiation operations which exhibit the usual arithmetic properties.

We define the addition map $+$: $\mathbf{N} \times \mathbf{N} \rightarrow \mathbf{N}$ by its unique transpose map $\hat{+} : \mathbf{N} \rightarrow \mathbf{N}^{\mathbf{N}}$, which is granted by cartesian closedness, such that the following diagram

$$\begin{array}{ccccc}
 & & \mathbf{N} & \xrightarrow{\text{suc}} & \mathbf{N} \\
 & \nearrow 0 & \downarrow \hat{+} & & \downarrow \hat{+} \\
 1 & & \mathbf{N}^{\mathbf{N}} & \xrightarrow{\text{suc}^{\mathbf{N}}} & \mathbf{N}^{\mathbf{N}} \\
 & \searrow id_{\mathbf{N}} & & &
 \end{array}$$

commutes. It follows that the diagram

$$\begin{array}{ccccc}
 & & \mathbf{N} \times \mathbf{N} & \xrightarrow{\text{suc} \times id_{\mathbf{N}}} & \mathbf{N} \times \mathbf{N} \\
 & \nearrow 0 \times id_{\mathbf{N}} & \downarrow + & & \downarrow + \\
 \mathbf{N} & & \mathbf{N} & \xrightarrow{\text{suc}} & \mathbf{N} \\
 & \searrow id_{\mathbf{N}} & & &
 \end{array}$$

is equivalent to the first. That is, $0 + p = p$ and $\text{suc}(p) + q = \text{suc}(p + q)$.

Likewise we define the multiplication and exponentiation maps, $\times : \mathbf{N} \times \mathbf{N} \rightarrow \mathbf{N}$ and $\text{ind} : \mathbf{N} \times \mathbf{N} \rightarrow \mathbf{N}$ respectively, by their respective unique exponential transpose maps such that

$$\begin{array}{ccccccc}
 1 & \xrightarrow{0} & \mathbf{N} & \xrightarrow{\quad \quad \quad \text{suc} \quad \quad \quad} & \mathbf{N} \\
 \downarrow 0 & & \downarrow \hat{\times} & & \downarrow \hat{\times} \\
 \mathbf{N} & \xrightarrow{\mathbf{N}^{\mathbf{N} \rightarrow 1}} & \mathbf{N}^{\mathbf{N}} & \xrightarrow{id_{\mathbf{N}^{\mathbf{N}}} \times id_{\mathbf{N}}} & \mathbf{N}^{\mathbf{N}} \times \mathbf{N}^{\mathbf{N}} \cong (\mathbf{N} \times \mathbf{N})^{\mathbf{N}} & \xrightarrow{+^{\mathbf{N}}} & \mathbf{N}^{\mathbf{N}}
 \end{array}$$

and,

$$\begin{array}{ccccccc}
 1 & \xrightarrow{0} & \mathbf{N} & \xrightarrow{\quad \quad \quad \text{suc} \quad \quad \quad} & \mathbf{N} \\
 \downarrow \text{suc}(0) & & \downarrow \hat{\text{ind}} & & \downarrow \hat{\text{ind}} \\
 \mathbf{N} & \xrightarrow{\mathbf{N}^{\mathbf{N} \rightarrow 1}} & \mathbf{N}^{\mathbf{N}} & \xrightarrow{id_{\mathbf{N}^{\mathbf{N}}} \times id_{\mathbf{N}}} & \mathbf{N}^{\mathbf{N}} \times \mathbf{N}^{\mathbf{N}} \cong (\mathbf{N} \times \mathbf{N})^{\mathbf{N}} & \xrightarrow{\times^{\mathbf{N}}} & \mathbf{N}^{\mathbf{N}}
 \end{array}$$

commute.

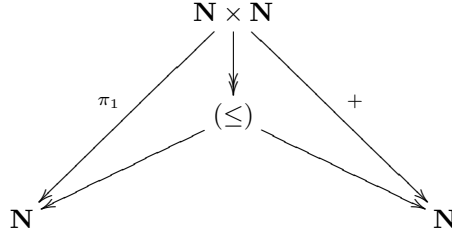
We have the isomorphism $\mathbf{N}^{\mathbf{N}} \times \mathbf{N}^{\mathbf{N}} \cong (\mathbf{N} \times \mathbf{N})^{\mathbf{N}}$ simply from the definition of an exponential object by homing in from $\mathbf{N}$. From demanding these two diagrams commute it is clear that we have the analogous properties of multiplication and exponentiation. That is $0 \times q = 0$, $\text{suc}(p) \times q = (p \times q) + q$ and $q^0 = 1$ with $q^{\text{suc}(p)} = q^p \times q$.

Finally, before we talk about ring objects in a topos, we want to consider a linear order on $\mathbf{N}$. When we do this it will make the process of identifying a linear order on the integers and rationals a more obvious process.

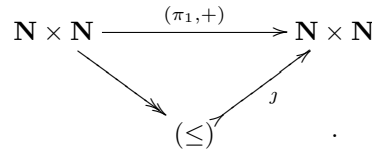
When one traditionally looks at the natural numbers there is a ‘natural’ sense of a linear ordering in which we say ‘ $a \leq b$ ’ if and only if $b = a + n$ for some n in $\mathbf{N}$. In a categorical sense we take the following diagram of how b and the pair (a, n) are related



We now take the image of this cone in $\mathbf{N} \times \mathbf{N}$ by an epi and then project out such that the following diagram



commutes. This product diagram can then of course be rewritten as



It is then the subobject $[(\leq), j]$ that yields us our linear order on $\mathbf{N}$.

3.2 Ring Objects in a Topos

We will now define what is meant by a ring object in a topos. This will enable us to recognize the familiar ring structure of the integers and the rationals when we define them. When we say ring here we mean a commutative ring equipped with a unit.

Definition Let $\mathcal{E}$ be a topos (alternatively a category with finite limits). A ring object in $\mathcal{E}$ (also called a ring in $\mathcal{E}$) is an object $\mathcal{R} \in \text{Ob } \mathcal{E}$ together with five morphisms

$$\mathbf{1} \begin{array}{c} \xrightarrow{0} \\ \xrightarrow{1} \end{array} \mathcal{R} \begin{array}{c} \xrightarrow{\quad \vee \quad} \\ \xrightarrow{\quad \bullet \quad} \end{array} \mathcal{R} \begin{array}{c} \xrightarrow{\quad + \quad} \\ \xrightarrow{\quad \bullet \quad} \end{array} \mathcal{R} \times \mathcal{R}$$

such that the following diagrams²,

$$\begin{array}{ccc} \mathcal{R} \times \mathcal{R} & \begin{array}{c} \xleftarrow{0 \times id_{\mathcal{R}}} \\ \xrightarrow{1 \times id_{\mathcal{R}}} \end{array} & \mathcal{R} \begin{array}{c} \xrightarrow{id_{\mathcal{R}} \times 0} \\ \xrightarrow{id_{\mathcal{R}} \times 1} \end{array} \mathcal{R} \times \mathcal{R} \\ & \searrow \quad \downarrow \quad \swarrow & \\ & \mathcal{R} & \end{array} \quad \begin{array}{ccc} \mathcal{R} \times \mathcal{R} & \xrightarrow{id_{\mathcal{R}} \times -} & \mathcal{R} \times \mathcal{R} \\ \delta \uparrow & & \downarrow + \\ \mathcal{R} & \xrightarrow{!} & \mathbf{1} \xrightarrow{0} \mathcal{R} \end{array}$$

$$\begin{array}{ccc} \mathcal{R} \times \mathcal{R} \times \mathcal{R} & \begin{array}{c} \xrightarrow{+ \times id_{\mathcal{R}}} \\ \xrightarrow{\bullet \times id_{\mathcal{R}}} \end{array} & \mathcal{R} \times \mathcal{R} \\ id_{\mathcal{R}} \times + \downarrow & id_{\mathcal{R}} \times \bullet \downarrow & \bullet \downarrow + \\ \mathcal{R} \times \mathcal{R} & \xrightarrow{\quad \bullet \quad} & \mathcal{R} \end{array} \quad \begin{array}{ccc} \mathcal{R} \times \mathcal{R} \cong \mathcal{R} \times \mathcal{R} & \xrightarrow{\quad + \quad} & \mathcal{R} \\ + \downarrow & & \downarrow id_{\mathcal{R}} \\ \mathcal{R} & \xrightarrow{id_{\mathcal{R}}} & \mathcal{R} \end{array}$$

$$\begin{array}{ccc} \mathbf{1} \times \mathcal{R} & \xrightarrow{\bullet(0, -)} & \mathcal{R} \\ ! \downarrow & \searrow 0 & \\ \mathbf{1} & & \end{array} \quad \begin{array}{ccc} \mathcal{R} \times \mathcal{R} \times \mathcal{R} & \xrightarrow{id_{\mathcal{R}} \times +} & \mathcal{R} \times \mathcal{R} \\ \delta \times id_{\mathcal{R}} \downarrow & & \downarrow \bullet \\ \mathcal{R} \times \mathcal{R} \times \mathcal{R} \times \mathcal{R} & & \\ \bullet \times \bullet \downarrow & & \\ \mathcal{R} \times \mathcal{R} & \xrightarrow{\quad + \quad} & \mathcal{R} \end{array}$$

commute.

It is clear that these commutative diagrams directly parallel the usual axioms given in most texts for a commutative ring. This idea of internal ring objects in a topos can be taken much further however this is all we shall require.

²where δ is the diagonal map, i.e. $x \rightarrow (x, x)$

3.3 The Integers

We want to form the integers as equivalence classes of pairs of natural numbers. Traditionally one constructs the integers as pairs of naturals modulo the equivalence relation $\sim$ defined by $(m, n) \sim (p, q)$ whenever $m + q = p + n$. Naturally we take a categorical analogue to this by considering the pullback

$$\begin{array}{ccc} \hat{\mathbf{H}} & \longrightarrow & \mathbf{N} \times \mathbf{N} \\ \downarrow & & \downarrow + \\ \mathbf{N} \times \mathbf{N} & \xrightarrow{+} & \mathbf{N} \end{array}$$

Where by $\hat{\mathbf{H}}$ we mean the object whose elements are quadruples (m, q, p, n) for which $m + q = p + n$. The object we seek is now the coequalizer

$$\hat{\mathbf{H}} \rightrightarrows \mathbf{N} \times \mathbf{N} \longrightarrow \mathbf{Z}.$$

Where $\mathbf{Z}$ is the ‘integer object’ and can be viewed as $(\mathbf{N} \times \mathbf{N})/\hat{\mathbf{H}}$.

When one traditionally constructs the integers and the rationals they naturally incur a commutative ring structure. It is very nice that the categorical construction also admits a similar structure. By an integer we mean an equivalence class of natural number elements $[x, y]$ so we define the addition and multiplication maps $+, \times : \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ by

$$[p, q] + [m, n] = [p +_{\mathbf{N}} m, q +_{\mathbf{N}} n] \text{ and,}$$

$$[p, q] \times [m, n] = [(p \times_{\mathbf{N}} m) +_{\mathbf{N}} (q \times_{\mathbf{N}} n), (p \times_{\mathbf{N}} n) +_{\mathbf{N}} (q \times_{\mathbf{N}} m)].$$

Where $+_{\mathbf{N}}$ and $\times_{\mathbf{N}}$ are the natural number addition and multiplication defined before. It is then a simple exercise to see that the ring object diagrams commute.

One can also define a linear order on $\mathbf{Z}$ such that it is an *ordered* ring in $\mathcal{E}$. Returning to the above where we took the pull back in order to get the relation on $\mathbf{N} \times \mathbf{N}$ as

$$\hat{\mathbf{H}} \rightrightarrows \mathbf{N} \times \mathbf{N},$$

we see that we can use the existence of powerobjects in an elementary topos. That is, there exists a unique $\hat{r}$ such that we can form the following factorization of $\hat{r}$

$$\begin{array}{ccccc} \mathbf{N} & \xrightarrow{(id_{\mathbf{N}}, 0)} & \mathbf{N} \times \mathbf{N} & \xrightarrow{\hat{r}} & \wp(\mathbf{N} \times \mathbf{N}) \\ & & \searrow & \nearrow & \\ & & \mathbf{Z} & & \end{array},$$

yielding the inclusion $\mathbf{N} \hookrightarrow \mathbf{Z}$.

We can now use this inclusion to construct the following alternative integer addition

$$\begin{array}{ccc} \mathbf{Z} \times \mathbf{N} & \xrightarrow{\quad} & \mathbf{Z} \times \mathbf{Z} \\ & \searrow +' & \downarrow + \\ & & \mathbf{Z} \end{array}$$

Once again we now take image of $\mathbf{Z} \times \mathbf{N}$ in $\mathbf{Z} \times \mathbf{Z}$ by an epi and then project out to obtain the following diagram

$$\begin{array}{ccc} & \mathbf{Z} \times \mathbf{N} & \\ \pi_1 \swarrow & \downarrow & \searrow +' \\ & (\leq) & \\ \swarrow & & \searrow \\ \mathbf{Z} & & \mathbf{Z} \end{array}$$

which we can now rewrite as,

$$\begin{array}{ccc} \mathbf{Z} \times \mathbf{N} & \xrightarrow{(\pi_1, +')} & \mathbf{Z} \times \mathbf{Z} \\ & \searrow & \nearrow j \\ & (\leq) & \end{array}$$

Just as in the case of the naturals we now have the linear order we sought as the subobject $[(\leq), j]$ of $\mathbf{Z} \times \mathbf{Z}$. We now see that the integers object is an internal ordered ring object of the topos $\mathcal{E}$ analogous to what is expected because of the ‘classical’ case.

3.4 The Rationals

Once the construction of the integers is discovered it is a similar process to obtain the rational numbers. Where as one traditionally constructs the rationals as pairs of integers, we would like to avoid the use of ‘A can not be equal to B’ statements. Hence we consider the pullback

$$\begin{array}{ccc} \hat{\mathbf{K}} & \xrightarrow{\quad} & \mathbf{Z} \times \mathbf{N} \\ \downarrow & & \downarrow \times (id_{\mathbf{Z}} \times suc) \\ \mathbf{Z} \times \mathbf{N} & \xrightarrow{\quad \times (id_{\mathbf{Z}} \times suc) \quad} & \mathbf{Z} \end{array}$$

where $\hat{\mathbf{K}}$ is the object of all (m, q, p, n) such that $m \times (q + 1) = p \times (n + 1)$. We therefore define the rationals as the coequaliser

$$\hat{\mathbf{K}} \rightrightarrows \mathbf{Z} \times \mathbf{N} \longrightarrow \mathbf{Q},$$

where $\mathbf{Q}$ is the ‘rationals object’ and can be viewed as $(\mathbf{Z} \times \mathbf{N})/\hat{\mathbf{K}}$.

Just as for the integers object we can define addition and multiplication maps for the rationals. We have constructed the rationals as the equivalence classes of an integer paired with a natural and as such we define the maps $+, \times : \mathbf{Q} \times \mathbf{Q} \rightarrow \mathbf{Q}$ by

$$[m, n] + [p, q] = [(m \times_{\mathbf{Z}} [\text{suc}(q), 0]_{\mathbf{Z}}) +_{\mathbf{Z}} (p \times_{\mathbf{Z}} [\text{suc}(n), 0]_{\mathbf{Z}}), (n \times_{\mathbf{N}} q) +_{\mathbf{N}} (n +_{\mathbf{N}} q)]$$

$$[m, n] \times [p, q] = [m \times_{\mathbf{Z}} p, (n \times_{\mathbf{N}} q) +_{\mathbf{N}} (n +_{\mathbf{N}} q)]$$

which satisfy the commutative diagrams in the definition of an internal ring object.

We now wish to construct a linear order on $\mathbf{Q}$. When we constructed the linear order on the integers in the previous section it was relatively straight forward as we had a grounded object of ‘positive’ integers; namely $\mathbf{N}$. However here we are posed with a minor problem as the rationals object (inclusion wise) is ‘bigger’ than the integers. Thus, we construct the object of ‘non negative’ rationals, $\mathbf{Q}^{\geq 0}$, such that the following diagram

$$\begin{array}{ccc} \mathbf{N} \times \mathbf{N} & \xrightarrow{\quad} & \mathbf{Z} \times \mathbf{N} \\ & \searrow & \searrow \\ & \mathbf{Q}^{\geq 0} & \xrightarrow{\quad} \mathbf{Q} \end{array}$$

commutes. Looked at from a purely ‘classical’ point of view, the set $\mathbf{Q}^{\geq 0}$ is quite literally the non-strickly positive rational numbers.

We can now proceed as before in that we take the image of $\mathbf{Q} \times \mathbf{Q}^{\geq 0}$ in $\mathbf{Q} \times \mathbf{Q}$ by an epi and then project

$$\begin{array}{ccccc} & & \mathbf{Q} \times \mathbf{Q}^{\geq 0} & & \\ & \swarrow \pi_1 & \downarrow & \searrow +' & \\ & & (\leq) & & \\ & \swarrow & & \searrow & \\ \mathbf{Q} & & & & \mathbf{Q}^{\geq 0} \end{array}$$

Once again we can rewrite this as

$$\begin{array}{ccc} \mathbf{Q} \times \mathbf{Q}^{\geq 0} & \xrightarrow{(\pi_1, +')} & \mathbf{Q} \times \mathbf{Q} \\ & \searrow & \nearrow j \\ & (\leq) & \end{array}$$

Thus we now have the linear order we were seeking as the subobject $[(\leq), j]$ of $\mathbf{Q} \times \mathbf{Q}^{\geq 0}$. Once again, in analogy to the ‘classical’ case, we have that $\mathbf{Q}$ is an internal ordered ring object of the topos $\mathcal{E}$.

3.5 The Reals by Dedekind Cuts

Classically there are two ways to construct the reals from the rationals. The first and most commonly referenced is the equivalence classes of Cauchy sequences method. The second, and the one we shall use, is by the construction of Dedekind cuts. In the language of toposes, the Dedekind reals are obtained by the conjunction in some way of a number of relations paralleling the usual logic-theoretic statements. For revision's sake we state the usual five conditions on a pair of subsets (L, U) of $\mathbb{Q}$ which are:

- (i) $\exists q \in \mathbb{Q}(q \in L), \exists q \in \mathbb{Q}(q \in U)$;
- (ii) $\forall q, r \in \mathbb{Q}(q < r \wedge r \in L \Rightarrow q \in L)$,
 $\forall q, r \in \mathbb{Q}(r < q \wedge r \in U \Rightarrow q \in U)$;
- (iii) $\forall q \in \mathbb{Q}(q \in L \Rightarrow \exists r \in \mathbb{Q}(r \in L \wedge q < r))$,
 $\forall q \in \mathbb{Q}(q \in U \Rightarrow \exists r \in \mathbb{Q}(r \in U \wedge r < q))$;
- (iv) $\forall q, r \in \mathbb{Q}(q < r \Rightarrow (q \in L \vee r \in U))$;
- (v) $L \cap U = \emptyset$

We will now translate these into certain subobjects of $\wp\mathbb{Q} \times \wp\mathbb{Q}$.

3.5.1 A Logic Analogy

In order to construct the subobjects we seek we require five other relations which parallel the usual logic tools; that is $\in$, $\cap$, $\subseteq$, $<$ and $\Rightarrow$.

The first one we look at is the parallel to the ' $\in$ ' or the 'is a member of' logic element. By simple application of the definition of relations, subobject classifier and then exponentials we have

$$Rel_{\mathcal{E}}(\mathbb{Q}, -) \cong Sub_{\mathcal{E}}(\mathbb{Q} \times -) \cong Hom_{\mathcal{E}}(\mathbb{Q} \times -, \Omega) \cong Hom_{\mathcal{E}}(-, \Omega^{\mathbb{Q}}).$$

However we see that $\Omega^{\mathbb{Q}} \cong \wp\mathbb{Q}$. So we also have

$$Rel_{\mathcal{E}}(\mathbb{Q}, -) \cong Hom_{\mathcal{E}}(-, \wp\mathbb{Q})$$

and thus, by Yoneda's Lemma, we set ' $-$ ' to $\wp\mathbb{Q}$ and run through by $id_{\wp\mathbb{Q}} \cong id_{\Omega^{\mathbb{Q}}}$ to obtain

$$id_{\Omega^{\mathbb{Q}}} \iff \mathbb{Q} \xrightarrow{\overline{\epsilon}_{\mathbb{Q}}} \wp\mathbb{Q}.$$

That is the identity map $id_{\Omega^{\mathbb{Q}}}$ corresponds to the relation $\overline{\epsilon}_{\mathbb{Q}} : \mathbb{Q} \rightarrow \wp\mathbb{Q}$. Since we have all exponentials we necessarily have

$$\begin{array}{ccc} \Omega^{\mathbb{Q}} \times \mathbb{Q} & \xrightarrow{eval} & \Omega \\ \uparrow id_{\Omega^{\mathbb{Q}}} \times id_{\mathbb{Q}} & \nearrow x \overline{\epsilon}_{\mathbb{Q}} & \\ \Omega^{\mathbb{Q}} \times \mathbb{Q} & & \end{array},$$

where $\chi_{\overline{\in \mathbf{Q}}} \rightarrow$ is the characteristic map of $\overline{\in \mathbf{Q}}$. Taking the pullback of $eval$ along $true$ yields the desired relation as subobject of $\Omega^{\mathbf{Q}} \times \mathbf{Q}$ in

$$\begin{array}{ccc} \in \mathbf{Q} & \xrightarrow{(\epsilon_2, \epsilon_1)} & \Omega^{\mathbf{Q}} \times \mathbf{Q} \\ \downarrow ! & & \downarrow eval \\ \mathbf{1} & \xrightarrow{true} & \Omega. \end{array}$$

We will also require an intersection operation which is relatively simple as we have all pullbacks in a topos. To say that $x \in H \cap K$ is to say that $xR(H, K)$ if and only if $x \in H$ and $x \in K$. So we consider the relations

$$Rel_{\mathcal{E}}(\wp \mathbf{Q} \times \wp \mathbf{Q}, \mathbf{Q}) \cong Hom_{\mathcal{E}}(\wp \mathbf{Q} \times \wp \mathbf{Q}, \wp \mathbf{Q}).$$

Thus we are looking for a morphism $\cap : \wp \mathbf{Q} \times \wp \mathbf{Q} \rightarrow \wp \mathbf{Q}$. By Yoneda's Lemma this is in correspondence to a natural transformation (for some X),

$$\begin{aligned} & Hom_{\mathcal{E}}(X, \wp \mathbf{Q} \times \wp \mathbf{Q}) \rightarrow Hom_{\mathcal{E}}(X, \wp \mathbf{Q}) \\ \iff & Hom_{\mathcal{E}}(X, \wp \mathbf{Q}) \times Hom_{\mathcal{E}}(X, \wp \mathbf{Q}) \rightarrow Hom_{\mathcal{E}}(X, \wp \mathbf{Q}) \\ \iff & Sub_{\mathcal{E}}(\mathbf{Q}) \times Sub_{\mathcal{E}}(\mathbf{Q}) \rightarrow Sub_{\mathcal{E}}(\mathbf{Q}) \end{aligned}$$

Thus for two subobjects R and S we define the intersection map by the pull back

$$\begin{array}{ccc} R \cap S & \xrightarrow{\quad} & S \\ \downarrow & & \downarrow \\ R & \xrightarrow{\quad} & \wp \mathbf{Q}. \end{array}$$

To say that $H \subseteq K$ is to say that $H \cap K = H$. Since we have the intersection map it is now straight forward to define the parallel as the equalizer

$$(\subseteq) \longrightarrow \wp \mathbf{Q} \times \wp \mathbf{Q} \xrightarrow[\pi_1]{\cap} \wp \mathbf{Q}.$$

In section 3.4 we fleshed out the rational numbers object and showed it was an ordered ring object. However the order we defined had a trichotomy nature which does not respect the requirements demanded of a Dedekind cut. What we require is a linear ordering without the 'zero' difference possibility. We thus consider the pullback

$$\begin{array}{ccccc} \hat{K}' & \xrightarrow{\quad} & \mathbf{N} \times \mathbf{N} & & \\ \downarrow & & \downarrow suc \times suc & & \\ & & \mathbf{N} \times \mathbf{N} & \xrightarrow{\times} & \\ \mathbf{N} \times \mathbf{N} & \xrightarrow{suc \times suc} & \mathbf{N} \times \mathbf{N} & \xrightarrow{\times} & \mathbf{N}. \end{array}$$

We then have the non-negative, non-zero rationals as the coequalizer (with the accompanying inclusion into the rationals) in the following diagram

$$\begin{array}{ccccc} \hat{K}' & \rightrightarrows & \mathbf{N} \times \mathbf{N} & \longrightarrow & \mathbf{Q}^{>0} \\ & \searrow & \searrow & & \downarrow \\ & & \hat{K} & \rightrightarrows & \mathbf{Z} \times \mathbf{N} \longrightarrow \mathbf{Q}. \end{array}$$

Thus our analogue ordering is given by the subobject of $\mathbf{Q} \times \mathbf{Q}$ in

$$\begin{array}{ccc} \mathbf{Q} \times \mathbf{Q}^{>0} & \xrightarrow{(\pi_1, +')} & \mathbf{Q} \times \mathbf{Q} \\ & \searrow & \nearrow j \\ & (<) & \end{array} .$$

The last logic tool we require is perhaps the most important: implication. To say $cRa \Rightarrow cSb$, where R and S are relations, is to say $s(b) \subseteq r(a)$. That is, the set of all c 's related to b under S must be contained in the set of all c 's related to a under R . Categorically, we have two relations $R \rightrightarrows C \times A$ and $S \rightrightarrows C \times B$, such that by use of the definition of powerobjects we can consider the following diagram

$$\begin{array}{ccccc} & & (\subseteq) & & \\ B & \searrow s & \swarrow & \searrow & A \\ & \wp C & & \wp C & \end{array} .$$

Taking the double pullback (that is the pullback of the smaller diamonds along $\subseteq$ and then the pullback of the resultant) yields

$$\begin{array}{ccccc} & & R \downarrow S & & \\ & \swarrow & & \searrow & \\ B & \swarrow \beta & & \searrow \alpha & A \\ & \searrow & (\subseteq) & \swarrow & \\ & \wp C & & \wp C & \end{array} .$$

This relation $R \downarrow S \rightrightarrows B \times A$ is now read $b(R \downarrow S)a$ if and only if $(cRa \Rightarrow cSb)$. The ‘down arrow’ notation used here is Mac Lane’s notation for a comma category.

3.5.2 The Categorical Approach

Now that we have the required logic tools we can begin to construct the five relations (subobjects) that define a Dedekind cut.

Dedekind(i)

$$\exists q \in \mathbb{Q}(q \in L), \exists q \in \mathbb{Q}(q \in U)$$

This is simply the subobject

$$\begin{array}{ccc} \in_{\mathbb{Q}} \times \in_{\mathbb{Q}} & \xrightarrow{\quad} & \wp \mathbb{Q} \times \wp \mathbb{Q} \\ & \searrow \quad \nearrow & \\ & \mathcal{R}_{(i)} & \end{array} .$$

Dedekind(ii)

$$\begin{array}{l} \forall q, r \in \mathbb{Q}(q < r \wedge r \in L \Rightarrow q \in L) \\ \forall q, r \in \mathbb{Q}(r < q \wedge r \in U \Rightarrow q \in U) \end{array}$$

As shown in section 3.5.1 our logic relations $<_{\mathbb{Q}}$ and $\in_{\mathbb{Q}}$ as subobjects are equivalent to morphisms $<_{\mathbb{Q}}: \mathbb{Q} \rightarrow \mathbb{Q}$ and $\in_{\mathbb{Q}}: \mathbb{Q} \rightarrow \wp \mathbb{Q}$ respectively. Thus we shall build up our relation piece by piece starting with,

$$H_1 = \in_{\mathbb{Q}} \circ <_{\mathbb{Q}}: \mathbb{Q} \longrightarrow \wp \mathbb{Q}$$

such that to say “ $(q, L) \in H_1$ ” is to say “ $\exists r \in \mathbb{Q}$ such that $q < r \in L$ ”. We require $H_1 \Rightarrow \in_{\mathbb{Q}}$ and so we use the implication device derived in 3.5.1 giving

$$H_1 \downarrow \in_{\mathbb{Q}} = \{(L, L') \in \wp \mathbb{Q} \times \wp \mathbb{Q} \mid \forall q \in \mathbb{Q}, (q, L) \in H_1 \Rightarrow q \in L'\}.$$

We shall force $L = L'$ thus yielding the first component of the desired relation. This is done by taking the following pullback (where δ is the diagonalization morphism)

$$\begin{array}{ccc} H_1 & \xrightarrow{\quad} & H_1 \downarrow \in_{\mathbb{Q}} \\ \downarrow & & \downarrow \\ \wp \mathbb{Q} & \xrightarrow{\quad \delta \quad} & \wp \mathbb{Q} \times \wp \mathbb{Q} . \end{array}$$

We will now construct the upper cut version of this subobject by first using $\ni_{\mathbb{Q}}: \wp \mathbb{Q} \rightarrow \mathbb{Q}$ (which is simply the corresponding morphism to $\in_{\mathbb{Q}}$ in the opposite order) in order to have,

$$K_1 = <_{\mathbb{Q}} \circ \ni_{\mathbb{Q}}: \wp \mathbb{Q} \longrightarrow \mathbb{Q}$$

such that to say “ $(U, q) \in K_1$ ” is to say “ $\exists r \in \mathbb{Q}$ such that $U \ni r < q$ ”. We require $K_1 \Rightarrow \in_{\mathbb{Q}}$ so, in keeping with the method of construction in 3.5.1, we have $K_1^{op} \downarrow \in_{\mathbb{Q}}$ defined as

$$K_1^{op} \downarrow \in_{\mathbb{Q}} = \{(L, L') \in \wp \mathbb{Q} \times \wp \mathbb{Q} \mid \forall q \in \mathbb{Q}, (U, q) \in H_1 \Rightarrow q \in U'\}.$$

Once again we force $U = U'$ by taking the pullback

$$\begin{array}{ccc}
 K_2 & \xrightarrow{\quad} & K_1^{op} \downarrow \in_{\mathbf{Q}} \\
 \downarrow & & \downarrow \\
 \wp \mathbf{Q} & \xrightarrow[\delta]{} & \wp \mathbf{Q} \times \wp \mathbf{Q}.
 \end{array}$$

Thus we obtain our desired subobject (relation) by

$$\mathfrak{R}_{(ii)} = H_2 \times K_2 \xrightarrow{\quad} \wp \mathbf{Q} \times \wp \mathbf{Q}.$$

Dedekind(iii)

$$\forall q \in \mathbf{Q} (q \in L \Rightarrow \exists r \in \mathbf{Q} (r \in L \wedge q < r))$$

$$\forall q \in \mathbf{Q} (q \in U \Rightarrow \exists r \in \mathbf{Q} (r \in U \wedge r < q))$$

Just as before we have the the two relations $\in_{\mathbf{Q}}$ and $<_{\mathbf{Q}}$ and their equivalents $\ni_{\mathbf{Q}}$ and $>_{\mathbf{Q}}$ in the opposite order. We also use a very similar piecewise construction starting with

$$M_1 = >_{\mathbf{Q}} \circ \ni_{\mathbf{Q}} : \wp \mathbf{Q} \longrightarrow \mathbf{Q}$$

such that to say “ $(L, q) \in M_1$ ” is to say “ $\exists r \in \mathbf{Q}$ such that $L \ni r$ with $r < q$ ”. We require $\in_{\mathbf{Q}} \Rightarrow M_1$ and so we consider

$$\in_{\mathbf{Q}} \downarrow M_1^{op} = \{(L, L') \in \wp \mathbf{Q} \times \wp \mathbf{Q} \mid \forall q \in \mathbf{Q}, q \in L \Rightarrow L' \ni r, r > q\}.$$

Just as in (ii) we take the pullback of $\in_{\mathbf{Q}} \downarrow M_1^{op}$ along δ (forcing $L = L'$) to yield

$$\begin{array}{ccc}
 M_2 & \xrightarrow{\quad} & \in_{\mathbf{Q}} \downarrow M_1^{op} \\
 \downarrow & & \downarrow \\
 \wp \mathbf{Q} & \xrightarrow[\delta]{} & \wp \mathbf{Q} \times \wp \mathbf{Q}.
 \end{array}$$

Just as before we build the relation on the upper cuts in a similar fashion starting with

$$N_1 = <_{\mathbf{Q}} \circ \ni_{\mathbf{Q}} : \wp \mathbf{Q} \longrightarrow \mathbf{Q}$$

such that to say “ $(U, q) \in N_1$ ” is to say “ $\exists r \in \mathbf{Q}$ such that $L \ni r$ with $r < q$ ”.

We require $\in_{\mathbf{Q}} \Rightarrow N_1$, so using the construction from 3.5.1, we have

$$\in_{\mathbf{Q}} \downarrow N_1^{op} = \{(U, U') \in \wp \mathbf{Q} \times \wp \mathbf{Q} \mid \forall q \in \mathbf{Q}, q \in U \Rightarrow U' \ni r, r < q\}.$$

Once again we force $U = U'$ by taking the pullback of N_1 along δ such that we have

$$\begin{array}{ccc} N_2 & \xrightarrow{\quad} & \in_{\mathbf{Q}} \downarrow N_1^{op} \\ \downarrow & & \downarrow \\ \wp \mathbf{Q} & \xrightarrow[\delta]{} & \wp \mathbf{Q} \times \wp \mathbf{Q}. \end{array}$$

Thus the relation we seek as a subobject of $\wp \mathbf{Q} \times \wp \mathbf{Q}$ is simply

$$\mathfrak{R}_{(ii)} = M_2 \times M_2 \xrightarrow{\quad} \wp \mathbf{Q} \times \wp \mathbf{Q}.$$

Dedekind(iv)

$$\forall q, r \in \mathbb{Q}(q < r \Rightarrow (q \in L \vee r \in U))$$

Consider the two following relations

$$\mathbb{I}_1 = \in_{\mathbf{Q}} \times \mathbf{Q} \times \wp \mathbf{Q} \xrightarrow{i_1} \mathbf{Q} \times \mathbf{Q} \times \wp \mathbf{Q} \times \wp \mathbf{Q}$$

$$\mathbb{I}_2 = \mathbf{Q} \times \wp \mathbf{Q} \times \in_{\mathbf{Q}} \xrightarrow{i_2} \mathbf{Q} \times \mathbf{Q} \times \wp \mathbf{Q} \times \wp \mathbf{Q}$$

defined by

$$(q \in L, r, U) \longmapsto (q, r, L, U)$$

$$(q, L, r \in U) \longmapsto (q, r, L, U)$$

respectively. The ‘or’ on both of these relations is simply the disjoint union (coproduct) of the two subobjects for which we then image under an epi to yield

$$\begin{array}{ccc} \mathbb{I}_1 \amalg \mathbb{I}_2 & \xrightarrow{[i_1, i_2]} & \mathbf{Q} \times \mathbf{Q} \times \wp \mathbf{Q} \times \wp \mathbf{Q} \\ & \searrow & \nearrow \\ & \mathbb{I} & \end{array}.$$

Using the same naming scheme as in 3.5.1 we let $A = \mathbf{1}$, $B = \wp \mathbf{Q} \times \wp \mathbf{Q}$, $C = \mathbf{Q} \times \mathbf{Q}$ and $R = (<) \rightharpoonup \mathbf{Q} \times \mathbf{Q}^3$ such that

$$\mathfrak{R}_{(iv)} = R \downarrow \mathbb{I} \xrightarrow{\quad} \wp \mathbf{Q} \times \wp \mathbf{Q}.$$

³Since $\mathbf{Q} \times \mathbf{Q} \cong \mathbf{Q} \times \mathbf{Q} \times \mathbf{1}$

Dedekind(v)

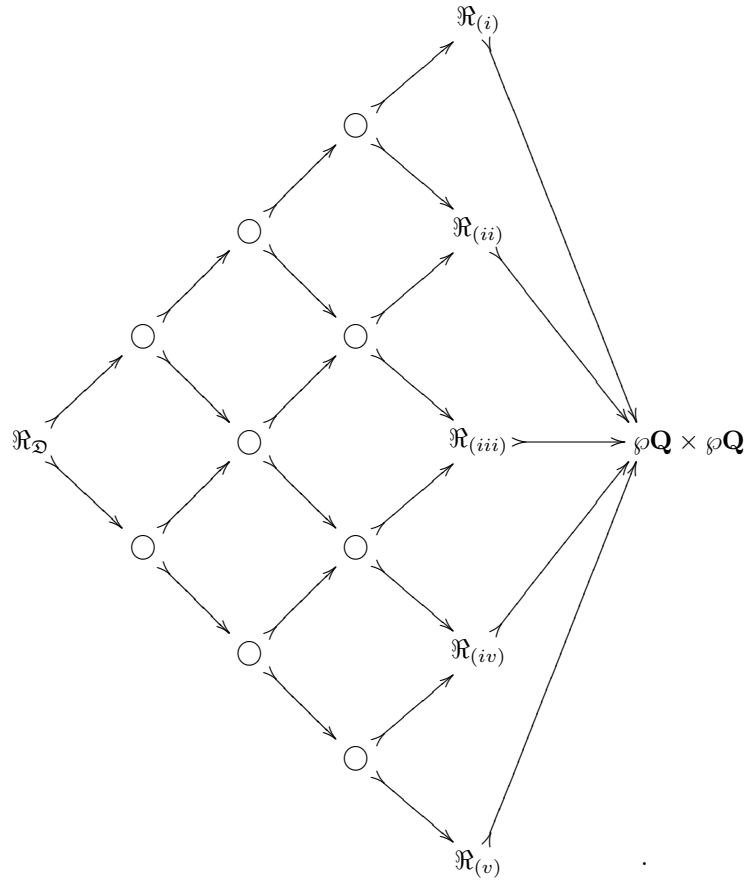
$$L \cap U = \phi$$

We have $\mathbf{0} \mapsto \mathbf{Q} \times \mathbf{1}$ for which we have the morphism $\phi : \mathbf{1} \rightarrow \wp \mathbf{Q}$ by definition. Thus our relation is obtained by the pullback of $\cap$ along ϕ in

$$\begin{array}{ccc} \mathfrak{R}_{(v)} & \xrightarrow{\quad ! \quad} & \mathbf{1} \\ \downarrow & & \downarrow \phi \\ \wp \mathbf{Q} \times \wp \mathbf{Q} & \xrightarrow{\quad \cap \quad} & \wp \mathbf{Q} . \end{array}$$

3.5.3 The Dedekind Reals Object

Now that we have the requisite five relations, expressed as subobjects of $\wp \mathbf{Q} \times \wp \mathbf{Q}$, one can now easily write down the Dedekind Reals object, $\mathfrak{R}_{\mathcal{D}}$, as the successive pullback or limit of the diagram



Chapter 4

The Filterpower Construction

4.1 Filters in Topology

In order to discuss what we will call an ultrapower construction, we require two basic definitions from topology.

Definition Let X be a set and F be a subset of $\wp X$. F is said to be a filter on X if

1. $F \neq \emptyset$ and $\emptyset \notin F$,
2. if $A, B \in F$ then $A \cap B \in F$,
3. if $A \in F$ and $A \subseteq B \subseteq X$ then $B \in F$.

Definition Let F be a filter on X . F is said to be an ultrafilter on X if we have the extra property that, for all $A \in \wp X$, either $A \in F$ or $\neg A \in F$.

4.1.1 Filters on $Sub(\mathbf{1})$

We can now talk about the concept of an ultrafilter on the subobjects of a terminal object in a topos. Let $\mathcal{E}$ be a topos. We have that ∇ is an ultrafilter on $Sub(\mathbf{1})$ if it admits the following twist on the original definition from topology,

1. $\mathbf{0} \notin \nabla^1$ and $\nabla \neq \emptyset$,
2. if $U, V \in \nabla$ then $U \times V \in \nabla$,
3. if $U \in \nabla$ and $U \subseteq V$ then $V \in \nabla$,
4. if $U \in Sub(\mathbf{1})$ then $U \in \nabla$ or $\neg U \in \nabla$.

¹where $\mathbf{0}$ is the initial object of $\mathcal{E}$ which is a subobject of every other object.

4.2 The Filterpower Construction

Definition Let $\mathcal{E}$ be a topos and ∇ be an ultrafilter on $Sub(\mathbf{1})$ of $\mathcal{E}$. We can form a new category $\mathcal{E}/\nabla$ where the objects of $\mathcal{E}/\nabla$ are the same as for $\mathcal{E}$ but where,

$$Hom_{\mathcal{E}/\nabla}(X, Y) = colim_{U \in \nabla} \mathcal{E}(X \times U, Y)$$

So a morphism $X \rightarrow Y$ in $\mathcal{E}/\nabla$ is an equivalence class $[U; f]$ where $U \in \nabla$ and $f : X \times U \rightarrow Y$.

To say $(U, f) \sim (V, g)$ means $\exists W \in \nabla$ with $W \subseteq U$, $W \subseteq V$ such that

$$\begin{array}{ccccc} & & X \times U & & \\ & id_X \times incl_U \nearrow & & f \searrow & \\ X \times W & & & & Y \\ & id_X \times incl_V \searrow & & g \nearrow & \\ & & X \times V & & \end{array}$$

commutes. Symmetry and reflexivity are given automatically where transitivity is given by

$$\begin{array}{ccccccc} & & & & X \times U & & \\ & & & & \nearrow & f \searrow & \\ & & X \times W & & & & \\ id_X \times \pi_1 \nearrow & & & & & & \\ X \times W \times W' & & & & X \times V & \xrightarrow{g} & Y \\ id_X \times \pi_2 \searrow & & & & \nearrow & g \nearrow & \\ & & X \times W' & & & & \\ & & \searrow & & X \times R & & \end{array} .$$

Finally we have an associative composition such that, given arrows $X \xrightarrow{[U, f]} Y$ and $Y \xrightarrow{[V, g]} Z$, we then have

$$X \xrightarrow{[U, f]} Y \xrightarrow{[V, g]} Z = [U \times V, g \circ (f \times id_V)]$$

That is,

$$\begin{array}{ccc} X \times U & \xrightarrow{f} & Y \\ Y \times V & \xrightarrow{g} & Z \\ X \times U \times V & \xrightarrow{f \times id_V} & Y \times V \xrightarrow{g} Z \end{array}$$

Proposition 4.2.1 (*SOE*) Given any collection of morphisms in $\mathcal{E}/\nabla$ of $f_1 : A_1 \times U_1 \rightarrow B_1, \dots, f_n : A_n \times U_n \rightarrow B_n$ we have that there exists $U \in \nabla$ where, $f'_1 : A_1 \times U \rightarrow B_1, \dots, f'_n : A_n \times U \rightarrow B_n$ with $(U_1, f_1) \sim (U, f'_1), \dots, (U_n, f_n) \sim (U, f'_n)$.

Proof Choose $U = U_1 \times \dots \times U_n$ such that,

$$\begin{array}{ccc} A_i \times U & \xrightarrow{f'_i} & B_i \\ & \searrow id_{A_i} \times \pi_i & \nearrow f_i \\ & A_i \times U_i & \end{array}$$

commutes. $\blacksquare$

Now we will define what shall be called the *projection functor* onto $\mathcal{E}/\nabla$. That is, a functor $P_\nabla : \mathcal{E} \rightarrow \mathcal{E}/\nabla$ such that it is bijective on objects and surjective on morphisms defined by $(f : A \rightarrow B) \mapsto ([1, f] : A \rightarrow B)$. The terminal object $\mathbf{1}$ is the natural choice here as it is the largest element in our ultrafilter.

Lemma 4.2.2 $\mathcal{E}/\nabla$ is finitely complete.

Proof It suffices to show the existence of a terminal object and that P_∇ preserves products and equalisers. Consider the morphisms from A to $\mathbf{1}$ in $\mathcal{E}/\nabla$:

$$Hom_{\mathcal{E}/\nabla}(A, \mathbf{1}) = colim_{U \in \nabla} Hom_{\mathcal{E}}(A \times U, \mathbf{1})$$

Clearly the right hand side is non-empty as it contains $[\mathbf{1}, !]$. Trivially we see that for every (U, f) we have $(U, f) \sim (\mathbf{1}, !)$ since

$$\begin{array}{ccc} & A \times \mathbf{1} & \\ id_A \times !_U \nearrow & & \searrow !_id \\ A \times U & & \mathbf{1} \\ id_A \times id_U \searrow & & \nearrow f \\ & A \times U & \end{array} .$$

Hence the terminal object of $\mathcal{E}/\nabla$ is simply that of $\mathcal{E}$.

We want to prove that $A \times B$ is the product of A and B in $\mathcal{E}/\nabla$. That is,

$$Hom_{\mathcal{E}/\nabla}(X, A \times B) \cong Hom_{\mathcal{E}/\nabla}(X, A) \times Hom_{\mathcal{E}/\nabla}(X, B)$$

So, given an $[V, f]$ onto A and $[V', g]$ onto B both based on X , we can invoke proposition *SOE* to have,

$$\begin{array}{ccc} X & \xrightarrow{[U, f]} & A \\ & \searrow [U, g] & \\ & & B \end{array}$$

in $\mathcal{E}/\nabla$ yielding the following diagram in $\mathcal{E}$

$$\begin{array}{ccc} & & A \\ & \nearrow f & \\ X \times U & & \\ & \searrow g & \\ & & B \end{array} \quad .$$

Now, $\mathcal{E}$ has all finite products so,

$$\begin{array}{ccccc} & & A & & \\ & \nearrow f & & \nwarrow \pi_1 & \\ X \times U & \xrightarrow{(f,g)} & A \times B & & \\ & \searrow g & & \swarrow \pi_2 & \\ & & B & & \end{array}$$

in $\mathcal{E}$. Under P_∇ we have a morphism $[U, (f, g)]$. For $A \times B$ to be the product in $\mathcal{E}/\nabla$ this must be the unique map. Suppose we have two maps $[V, h]$ and $[V, k]$ where,

$$\begin{array}{ccc} & & A \\ & \nearrow [1, \pi_1] & \\ X \xrightarrow{\quad [V, h] \quad} & A \times B & \\ & \searrow [1, \pi_2] & \\ & & B \end{array}$$

in $\mathcal{E}/\nabla$. This translates as two diagrams, one for each projection. Now, $V \in \nabla$ so in both cases we can take a W_1 and W_2 ‘further out’ with appropriate inclusions into V . We know $\mathcal{E}$ has all finite products and so must have $X \times W_1 \times W_2$. However we also have that $W_1 \times W_2$ is a subobject of $\mathbf{1}$ in the ultrafilter so the inclusions into V by composition must be unique. This is illustrated by

$$\begin{array}{ccccccc} & & & & X \times V & \xrightarrow{h} & A \times B \\ & & \nearrow id_X \times !incl & & \nearrow & & \searrow \pi_1 \\ X \times W_1 \times W_2 & \longrightarrow & X \times W_1 & & & & \\ & & \searrow & & \searrow & & \swarrow \pi_1 \\ & & & & X \times V & \xrightarrow{k} & A \times B \end{array}$$

and,

$$\begin{array}{ccccc}
 & & X \times V & \xrightarrow{h} & A \times B \\
 & \nearrow^{id_X \times !incl} & & & \searrow \pi_2 \\
 X \times W_1 \times W_2 & \longrightarrow & X \times W_2 & & \\
 & \searrow_{id_X \times !incl} & & & \nearrow \pi_2 \\
 & & X \times V & \xrightarrow{k} & A \times B
 \end{array}$$

We can now call on the uniqueness of $A \times B$ in $\mathcal{E}$ to give $h \circ (id_X \times !incl) = k \circ (id_X \times !incl)$ for both projections and hence $[V, h] = [V, k]$ in $\mathcal{E}/\nabla$.

The last item we are required to prove is the existence of equalisers. As it has been seen once already, we shall omit the verbose use of proposition *SOE* and simply begin by having already used it. That is, we want to find the equalizer of

$$A \begin{array}{c} \xrightarrow{[U, f]} \\ \xrightarrow{[U, g]} \end{array} B$$

in $\mathcal{E}/\nabla$. Viewed in $\mathcal{E}$ this is

$$A \times U \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B$$

which we know has the equalizer $(E \times U, h \times id_U)$. We can also see that $A \times U \times U \cong A \times U$ such that the composition $\cong \circ (h \times id_U) = h \times id_U$ is unique. That is, we have

$$\begin{array}{ccccc}
 E \times U & \xrightarrow{h \times id_U} & A \times U \times U & \xrightarrow{\cong} & A \times U \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \\
 & \searrow_{h \times id_U} & & &
 \end{array}$$

in $\mathcal{E}$. Hence

$$E \xrightarrow{[U, h \times id_U]} A \begin{array}{c} \xrightarrow{[U, f]} \\ \xrightarrow{[U, g]} \end{array} B$$

■

Theorem 4.2.3 $\mathcal{E}/\nabla$ is a topos.

Proof By lemma 3.2.2 we have that $\mathcal{E}/\nabla$ is finitely complete and that P_∇ preserves products and equalisers. All that is left is to show that $\mathcal{E}/\nabla$ is cartesian closed and has a subobject classifier.

Consider the diagram of $[U, f] : X \times Y \rightarrow Z$ from $\mathcal{E}/\nabla$ represented in $\mathcal{E}$,

$$X \times Y \times U \xrightarrow{f} Z$$

Here Z and $Y \times U$ have an exponential object such that we have

$$\begin{array}{ccc} Z^{Y \times U} \times Y \times U & \xrightarrow{eval} & Z \\ \uparrow !\hat{f} \times id_{Y \times U} & \nearrow f & \\ X \times Y \times U & & \end{array} .$$

We claim that $[U, !\hat{f} \times id_{Y \times U}]$ is the unique map making everything commute such that Z^Y is the exponential object in $\mathcal{E}/\nabla$. Suppose we have two morphisms $[U, h], [U, k] : X \times Y \rightarrow Z^Y \times Y$ making the exponential diagram in $\mathcal{E}/\nabla$ commute.² We can then take this to $\mathcal{E}$ to have the following

$$\begin{array}{ccccc} & & X \times Y \times U & & \\ id_X \times id_{Y \times U} \nearrow & & \downarrow h \times id_{Y \times U} & & \searrow f \\ X \times Y \times U & \xrightarrow{!\hat{f} \times id_{Y \times U}} & Z^{Y \times U} \times Y \times U & \xrightarrow{eval} & Z \\ id_X \times id_{Y \times U} \searrow & & \uparrow k \times id_{Y \times U} & & \nearrow f \\ & & X \times Y \times U & & \end{array} .$$

Invoking the uniqueness of $!\hat{f} \times id_{Y \times U}$ in $\mathcal{E}$ yields $(h \times id_{Y \times U}) \circ (id_X \times id_{Y \times U}) = (k \times id_{Y \times U}) \circ (id_X \times id_{Y \times U})$ and hence $[U, h] = [U, k]$ as required.

Lastly we are required to prove the existence of a subobject classifier. We claim that Ω is the subobject classifier in $\mathcal{E}/\nabla$. For this to be the case there must be an accompanying truth morphism. So, the right hand side of

$$Hom_{\mathcal{E}/\nabla}(\mathbf{1}, \Omega) = colim_{U \in \nabla} Hom_{\mathcal{E}}(\mathbf{1} \times U, \Omega),$$

is clearly non-empty since it contains $[\mathbf{1}, tr]$. However given any other applicable (U, f) we can see that $(U, f) \sim (\mathbf{1}, tr)$ since

$$\begin{array}{ccc} & \mathbf{1} \times U & \\ id_{\mathbf{1}} \times id_U \nearrow & & \searrow f \\ \mathbf{1} \times U & & \Omega \\ id_{\mathbf{1}} \times !_U \searrow & & \nearrow tr \\ & \mathbf{1} \times \mathbf{1} & \end{array} .$$

So we have a unique truth morphism tr . Next we consider a monomorphism $[U, m] : S \rightarrow X$ in $\mathcal{E}/\nabla$. We are required to prove that Ω together with $[\mathbf{1}, tr]$ give a unique classification morphism for $[U, m]$. So, in $\mathcal{E}/\nabla$ we have

$$\begin{array}{ccc} S & \xrightarrow{[U, m]} & X \\ \downarrow ! & & \\ \mathbf{1} & \xrightarrow{[\mathbf{1}, tr]} & \Omega \end{array}$$

²we have used proposition *SOE* here.

which in $\mathcal{E}$ is

$$\begin{array}{ccc} S \times U & \xrightarrow{m} & X \cong X \times \mathbf{1} \\ \downarrow ! & & \downarrow !\chi_m^U \\ \mathbf{1} & \xrightarrow{tr} & \Omega \end{array}$$

for which there is a unique morphism $!\chi_m^U$. Noting that $X \cong X \times \mathbf{1}$ we claim that $[\mathbf{1}, !\chi_m^U]$ is the unique morphism making the diagram in $\mathcal{E}/\nabla$ commute. Suppose that both $[\mathbf{1}, h], [\mathbf{1}, k] : X \rightarrow \Omega$ make the above square in $\mathcal{E}/\nabla$ commute. Taking this to $\mathcal{E}$ we have

$$\begin{array}{ccc} S \times U & \xrightarrow{m} & X \times \mathbf{1} \\ \downarrow ! & & \downarrow k \\ \mathbf{1} & \xrightarrow{tr} & \Omega \\ \uparrow ! & & \uparrow h \\ S \times U & \xrightarrow{m} & X \times \mathbf{1} \end{array}$$

By the uniqueness of χ_m in $\mathcal{E}$ we have that $hom = kom$ and hence $[\mathbf{1}, h] = [\mathbf{1}, k]$ as required. $\blacksquare$

This proof further shows that the projection functor $P_{\nabla} : \mathcal{E} \rightarrow \mathcal{E}/\nabla$ is in fact logical. That is, it preserves the underlying structure of $\mathcal{E}$. This will be very important in the next chapter when we construct the object of non-standard reals.

Chapter 5

Non-Standard Real Numbers

In this chapter we look at the global sections functor. We then give a particular example of the filterpower construction introduced in the previous chapter and how it can be used to get at the non-standard real numbers as originally introduced by A. Robinson in his book on non-standard analysis. We begin with a review of what it means to have a non-standard real number.

5.1 Review of Basic Analysis

In traditional real analysis one has a concept of a real number for which one may make statements like ‘for epsilon greater than 0’. However before there was a solid idea of real number as there is today (e.g. equivalence classes of Cauchy sequences) it was common to say that one was ‘infinitely close to x ’ for some value x written $x + dx$ where dx was this infinitesimal distance¹.

To give a nice definition for what it means to be an ‘infinitesimal’ we first need one of the first theorems encountered in basic real analysis.

Theorem 5.1.1 (*Archimedian Property*) *For any $x \in \mathbb{R}$ there exists a $n \in \mathbb{N}$ such that $x \leq n$.*

That is to say that the natural numbers are cofinal in the reals.

A non-standard real number is therefore defined as an element of a non-archimedian field of real numbers. Less precisely one can visualize them as numbers between zero and the ‘start’ of the reals² on a number line. When A. Robinson went on to write his book he developed an entire rigorous logic and calculus system for dealing with them.

¹This notation is still in wide use today by physicists

²and ‘after’ the $\infty \in \mathbb{N}$

5.2 Global Sections

The global sections functor we will define here will be used in the ultimate construction of the non-standard real numbers object in the next section. First we will consider sections.

Definition Let $\mathcal{C}$ be a category with $f : X \rightarrow Y$ and $g : Y \rightarrow X$ morphisms in $\mathcal{C}$ such that $f \circ g = id_Y$ then f is called a *retraction* of g and we say g is a *section* of f .

This of course can be read as g is a right inverse for f and f is a left inverse for g . The definition of a terminal object is such that given any other object in the category there exists a unique morphism into the terminal object. Knowing this, the use of the word ‘global’ in the following definition makes more sense.

Definition Let $\mathcal{C}$ be a category and $A \in ob\mathcal{C}$. The *set of global sections* of A is defined to be $Hom_{\mathcal{C}}(\mathbf{1}, A)$ where $\mathbf{1}$ is the terminal object of the category.

We can now define the *global sections functor* $\Gamma : \mathcal{C} \rightarrow \mathbf{Set}$ as the functor sending an object A in $\mathcal{C}$ to the set $\Gamma A = Hom_{\mathcal{C}}(\mathbf{1}, A)$ and the morphism $f : A \rightarrow A'$ to the function $\Gamma f : Hom_{\mathcal{C}}(\mathbf{1}, A) \rightarrow Hom_{\mathcal{C}}(\mathbf{1}, A')$ defined by $(\Gamma f)(a) = fa$.

5.3 Topos of Sequences of Sets

Definition The category $\mathbb{N}$ is defined to be the discrete category³ whose objects are the natural numbers⁴.

We can now introduce the topos, which enables the construction of the non-standard reals, as the functor category $Func[\mathbb{N}, \mathbf{Set}]$. We have already proved that this is a topos in 2.2.1 and it is clear each functor $F : \mathbb{N} \rightarrow \mathbf{Set}$ assigns to each natural number a set. Denoting $Func[\mathbb{N}, \mathbf{Set}]$ by $\mathcal{E}$ then we see that every $A \in ob\mathcal{E}$ can actually be viewed as a sequence of sets indexed over the natural numbers; that is $A = (A_n)$. A morphism (natural transformation) $f : A \rightarrow B$ is a sequence $f = (f_n)$ of functions $f_n : A_n \rightarrow B_n$.

We have a natural notion of the terminal object⁵ being the constant sequence at the singleton set (the terminal object in $\mathbf{Set}$) which we denote as $\mathbf{1}$ of $\mathcal{E}$. It is then clear that a subobject U ⁶ of $\mathbf{1}$ is a sequence U_n of sets such that $|U_n| \leq 1$. Thus any subobject $U = (U_n)$ can be identified with a subset of the natural numbers $\{n | U_n \neq \emptyset\}$. More over any conceivable subset of $\mathbb{N}$ can be formed. For example, the empty set is formed by taking the constant sequence at the empty set in $\mathbf{Set}$ (i.e. the initial object). Thus we make the observation that we have the isomorphism $Sub_{\mathcal{E}}(\mathbf{1}) \cong \wp\mathbb{N}$. That is, the subobjects of $\mathbf{1}$ form the boolean algebra of subsets of natural numbers in $\mathbb{N}$.

³That is the only morphisms are the identities

⁴We include 0 in our definition of the natural numbers

⁵(cartesian closedness)

⁶Using the same notation as chapter 4.

An ultrafilter ∇ on $\wp\mathbb{N}$ is an ultrafilter on the set $\mathbb{N}$. As per chapter 4 we have the projection functor $P_\nabla : \mathcal{E} \rightarrow \mathcal{E}/\nabla$ which is bijective on objects and surjective on morphisms. That is, the objects of $\mathcal{E}/\nabla$ are simply those of $\mathcal{E}$; sequences of sets.

Again by chapter 4 a morphism $[U; f] : A \rightarrow B$ in $\mathcal{E}/\nabla$ is an equivalence class of pairs (U, f) ⁷ where we say that $(U, f) \sim (V, g)$ if there exists a $W \in \nabla$ such that $W \subset U \cap V$ with $f_n = g_n$ for all $n \in W$.

As was shown in chapter 4 the projection functor $P_\nabla : \mathcal{E} \rightarrow \mathcal{E}/\nabla$ is logical. This is very nice as it allows the easy identification of key structures in the topos $\mathcal{E}/\nabla$. For example the subobject classifier of $\mathcal{E}/\nabla$ is simply the constant sequence at the 2-set; the subobject classifier in **Set**. Moreover the naturals, integers and rationals number objects in $\mathcal{E}/\nabla$ are the constant sequence at the set of naturals, integers and rationals (respectively) from **Set**. Thus the Dedekind reals object in $\mathcal{E}/\nabla$ is $[\mathbb{R}_\mathbb{D}]$; the constant sequence at $\mathbb{R}_\mathbb{D}$.

Before we talk of global sections there is one matter that needs attention. Suppose ∇ is a principal ultrafilter, that is, $\nabla = \{U \subset \mathbb{N} | k \in U\}$ for some fixed $k \in \mathbb{N}$. Let $A \in \text{ob}\mathcal{E}/\nabla$. We notice that every morphism $A \times U \rightarrow B$ in $\mathcal{E} = \text{Func}[\mathbb{N}, \mathbf{Set}]$ is equivalent to a unique morphism of the form $A \times K \rightarrow B$ where K_m is empty for all m not equal to k , and a singleton for $m = k$. Such a morphism $A \times K \rightarrow B$ is a function $A_k \rightarrow B_k$ ⁸. Thus we have that $\mathcal{E}/\nabla$ is equivalent to **Set** leading to a very uninteresting case.

Proposition 5.3.1 *There exist non-principal ultrafilters of $\mathbb{N}$.*

Proof Let $\mathcal{C}$ be the set of all cofinite⁹ subsets of $\mathbb{N}$. This is clearly a proper filter on $\mathbb{N}$. The union of any ascending chain of proper filters is a proper filter. By Zorn's lemma $\mathcal{C}$ is contained in a maximal proper filter (an ultrafilter). It is clear that this is a non-principal ultrafilter since for any k there exists the cofinite subset $(k + 1, \infty)$. ■

So, assuming that ∇ is not principal, we have the following proposition.

Proposition 5.3.2 *∇ contains no finite sets.*

Proof Shall show that ∇ does not contain any sets of cardinality $k < \infty$ by induction.

$k = 0$ $\emptyset \notin \nabla$

$k = 1$ $S = \{x\}$

If $\{x\} \in \nabla$ then the principal filter $\mathcal{F}_x$ generated by x is contained in ∇ . Thus, by maximality (of ultrafilters), we have $\mathcal{F}_x = \nabla$. Contradiction to non-principality.

Suppose true for some $k = i - 1$.

$k = i$ $S = \{x_1, \dots, x_i\} \in \nabla$.

By the $k = 1$ case we have $\{x_1\} \notin \nabla$. Since ∇ is an ultrafilter we have $\neg\{x_1\} \in \nabla$. So $\{x_2, \dots, x_i\} = S \cap \neg\{x_1\} \in \nabla$. Contradiction to the induction hypothesis since $|\{x_2, \dots, x_i\}| = i - 1$. ■

⁷ $U \in \nabla$ and $f = (f_n)$ where $f_n : A_n \rightarrow B_n$

⁸this is because $A \times \emptyset$ is empty and there is only one function from the empty set to B_m .

⁹finite complement

Given an object A of $\mathcal{E}/\nabla$ we have the global sections ΓA as the set of equivalence classes of elements of $A = (A_n)$. That is, if a and b are elements of A then $a \sim b$ when $\{n | a_n = b_n\} \in \nabla$. ΓA is called the *ultrapower* of the A_n 's modulo the ultrafilter ∇ .

Proposition 5.3.3 *If $A \in \text{Ob } \mathcal{E}/\nabla$ is the constant sequence $[\bar{A}]$ at the set A then there exists a monomorphism $\eta_A : A \rightarrow \Gamma A$.*

Proof Define η_A by $a \mapsto [\bar{a}]$ where $\bar{a}_n = a$ for all $n \in \mathbf{N}$. Suppose $a, b \in A$ with $[\bar{a}] = [\bar{b}]$. Then

$$\{n \in \mathbf{N} | \bar{a}_n = \bar{b}_n\} \in \nabla$$

In particular $\{n \in \mathbf{N} | \bar{a}_n = \bar{b}_n\} \neq \emptyset$. So there exists a $k \in \mathbf{N}$ where $\bar{a}_k = \bar{b}_k$. Thus

$$a = \bar{a}_k = \bar{b}_k = b$$

and η_A is injective¹⁰. ■

Therefore the set of *non-standard reals* ${}^*\mathbf{R}$ is defined to be the *ultrapower* of the Dedekind reals object in $\mathcal{E}/\nabla$, that is ${}^*\mathbf{R} = \Gamma[\mathfrak{R}_{\mathfrak{D}}]$. The infinitesimals (and their reciprocals) are the elements of ${}^*\mathbf{R}$ not in the image of $\eta_{\mathfrak{R}_{\mathfrak{D}}}$ (as an inclusion in **Set**).

A standard real is an element of ${}^*\mathbf{R}$ of the form $(r, r, r, \dots)$ where $r \in \mathfrak{R}_{\mathfrak{D}}$. An example of a non-standard real is $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots) \in {}^*\mathbf{R}$. But is it an infinitesimal? To answer this question we need to define a suitable partial ordering on ${}^*\mathbf{R}$.

Definition If $a, b \in {}^*\mathbf{R}$ (the set of non-standard reals) then we say that $a \leq b$ if and only if $\{n \in \mathbf{N} | a_n \leq b_n\} \in \nabla$.

Thus $(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots)$ is indeed an infinitesimal since $\{n \in \mathbf{N} | \frac{1}{n+1} \leq r\} \in \nabla$ for any $r \in \mathfrak{R}_{\mathfrak{D}}$ (by the archimedian property and Proposition 5.3.2). Like-wise, the non-standard real $(1, 2, 3, \dots)$ is a number ‘after’ the point at infinity since for any $r \in \mathfrak{R}_{\mathfrak{D}}$ we have $\{n \in \mathbf{N} | r \leq n\} \in \nabla$.

In a similar way to the partial ordering above we can define addition and multiplication operations modulo a relation relative to the elements of ∇ .

Definition If $a, b \in {}^*\mathbf{R}$ then we say that

- $a + b = h \pmod{\nabla}$ provided $\{n | a_n + b_n = h_n\} \in \nabla$ and,
- $a \times b = k \pmod{\nabla}$ provided $\{n | a_n \times b_n = k_n\} \in \nabla$.

These operations together with the ordering defined above provide the necessary requirements for ${}^*\mathbf{R}$ to be an ordered field. Notice that there are no problems with zero divisors as one may initially think. To see this consider $(1, 0, 1, 0, \dots)$ and $(0, 1, 0, 1, \dots)$ and see that even though the product of the two is $[\bar{0}]$ due to maximality of ∇ one is $0 \pmod{\nabla}$ and the other is $1 \pmod{\nabla}$.

¹⁰in **Set** injections are precisely the monomorphisms

5.4 Non-Standard Analysis

The initial reason Abraham Robinson had to provide his original construction of the non-standard reals ${}^*\mathbf{R}$ was to be able to formulate a consistent model of non-standard analysis. Equipped with this reformulation one may then use non-standard arguments in areas such as physics or the study of partial differential equations. As such we shall present the proof of the non-standard version of the intermediate value theorem together with the required theory. The intermediate value theorem is a powerful tool in ‘elementary’ calculus and is used in the proof of the non-standard fundamental theorem of calculus.

5.4.1 Requisite Theory

In section 5.3 we showed how one may construct the non-standard reals object from a Dedekind reals object in a specific topos of sequences of sets. It now makes sense to elaborate on why we chose to call this object ${}^*\mathbf{R}$.

Definition Given a set A in **Set** we call *A the $*$ -transform of A defined by ${}^*A = \Gamma[\bar{A}]$; the global sections of the constant sequence $[\bar{A}]$ in $\mathcal{E}/\nabla$. Likewise given a morphism $f : A \rightarrow B$ in **Set** we can take the $*$ -transform of f defined as ${}^*f : ({}^\sigma A \rightarrow {}^\sigma B) \rightarrow ({}^\sigma B \subset {}^*B)$.

In this way one may have non-standard versions of the natural numbers object in ${}^*\mathbf{N}$. We then naturally have the induced monomorphism $\eta_{\mathbf{N}} : \mathbf{N} \rightarrow {}^*\mathbf{N}$. We call the image of $\eta_{\mathbf{N}}$ the standard copy of $\mathbf{N}$ in ${}^*\mathbf{N}$ and write it as ${}^\sigma\mathbf{N}$. Clearly an element of ${}^\sigma\mathbf{N}$ is a constant constant sequence at an element of $\mathbf{N}$.

Definition We call an element $(r_1, r_2, r_3, \dots)$ of ${}^*\mathbf{R}$ *finite* if $\sup_j \{r_j\} \leq m$ for some $m \in \mathbf{N}$. Similarly and we say $(k_1, k_2, k_3, \dots)$ is *infinitesimal* if $\inf_j \{r_j\} \leq \frac{1}{n}$ for every $n \in \mathbf{N}$.

We call $\mathbb{O}$ the set of finite elements of ${}^*\mathbf{R}$. Naturally we then have the set of infinitesimal elements o .

Proposition 5.4.1 (Stroyan) $\mathbb{O}$ is a totally ordered, Archimedean integral domain as a subring of ${}^*\mathbf{R}$. o is a maximal proper ideal, as well as an order ideal of $\mathbb{O}$.

For the sake of simplicity we omit the proof of proposition 5.4.1 although a mostly completed proof can be found in K.D. Stroyan’s ‘Introduction to the Theory of Infinitesimals’.

Definition The order homomorphism $\kappa : \mathbb{O} \rightarrow \mathbb{R}_{\mathfrak{D}}$ with kernel o is called the *standard part homomorphism*.

There is one more item we require before we have some more traditional calculus ideas. That is, some notation for when two non-standard reals are ‘close’ to one another. We say that $x, y \in {}^*\mathbf{R}$ are infinitely close to one another when $x - y \in o$. When this is the case we write $x \approx y$.

In order to begin discussing a non-standard model of calculus we ~~require a different~~¹¹ concept of continuity ~~which will take into account our infinitesimal calculus of~~ ${}^*\mathbf{R}$.

Definition Let $A \subset {}^*\mathbf{R}$ and $B \subset {}^*\mathbf{R}$. We say that the function $f : A \rightarrow B$ is *continuous* if and only if $f(x + dx) = f(x) + dz \approx f(x)$ where dx and dz are in o and x is in ${}^o\mathbf{R}$.

One will notice that this is a natural way to express the statement “small perturbations in the argument cause small perturbations in the value of the function”. Here we have tacitly used ‘small’ to mean an infinitesimal shift to the ‘left’ or the ‘right’.

Of course for the complete statement of the standard intermediate value theorem on $\mathbb{R}$ one is always equipped with the Heine-Borel theorem which says the compactness is the same as being closed and bounded. To get at the non-standard parallel to this we can define an order topology on ${}^*\mathbf{R}$ (relative to the ordering in section 5.3).

Definition We say that $K \subset {}^*\mathbf{R}$ is *compact* if and only if every open cover of K has a finite subcover.

The following theorem is stated with out proof due to its complexity though it can be found in K.D. Stroyan’s ‘Introduction to the Theory of Infinitesimals’.

Theorem 5.4.2 *Let $S \subset \mathbb{R}_D$ then S is compact if and only if every element of *S is infinitesimally near to a ‘standard’ element of ${}^o S$.*

Lastly we need to define the *sign function* as the function $sgn : {}^*\mathbf{R} \rightarrow \{-1, 0, 1\}$ such that

$$sgn(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}.$$

¹¹this is actually a misnomer as it can be shown that our definition is a sort of * -transform of the traditional epsilon-delta definition.

5.4.2 The Non-Standard Intermediate Value Theorem

Theorem 5.4.3 (*NsIVT*) *Let f be a continuous real $(\mathbb{R}_{\mathfrak{D}})$ valued function defined on the compact interval $[a, b]$ such that $f(a) \neq f(b)$, then f attains all values between $f(a)$ and $f(b)$ on (a, b) .*

Proof (Luxemburg) We are required to show that for all c such that $f(a) < c < f(b)$ there exists an x such that $f(x) = c$. With-out loss of generality we assume that $f(a) < 0$, $c = 0$ and $f(b) > 0$. If this is not the case then we can utilize the sign function to have

$$F(x) = \text{sgn}(f(b) - f(a))[f(x) - c]$$

which clearly¹² satisfies $F(a) < 0$ and $F(x) > 0$ with $F(x) = 0$ if and only if $f(x) = c$. Further we can shift our interval by a change in variables in

$$F(x) = F(a + x(b - a))$$

such that $F(0) = F(a)$ and $F(1) = F(b)$. Thus our problem is reduced to considering $f : [0, 1] \rightarrow \mathbb{R}_{\mathfrak{D}}$ where $f(a) < 0$, $c = 0$ and $f(b) > 0$.

We use a proof by contradiction. Suppose that f has no zeros. For all $n \in \mathbf{N}$, $n \geq 1$ we consider the non-zero telescoping product

$$\prod_{k=0}^{n-1} \frac{f((k+1)/n)}{f(k/n)} = \frac{f(1/n)}{f(0)} \frac{f(2/n)}{f(1/n)} \cdots \frac{f((n-1)/n)}{f((n-2)/n)} \frac{f(1)}{f((n-1)/n)} = \frac{f(1)}{f(0)}$$

for which the right hand side is negative. Since this is a product it implies that one of the terms¹³ in the product on the left hand side is also negative.

Let $a_n = k/n$ and $b_n = (k+1)/n$ such that for any $n \in \mathbf{N}$, $n \geq 1$ there exists numbers a_n and b_n such that $0 \leq a_n < b_n \leq 1$, $b_n - a_n = 1/n$ and $f(b_n)/f(a_n) < 0$.

If we take the $*$ -transform of the above line and choose n as an infinite $\omega \in {}^*\mathbf{N}$ we have $a_\omega, b_\omega \in {}^*[0, 1]$. Additionally,

$$b_\omega - a_\omega = \frac{k+1}{\omega} - \frac{k}{\omega} = \frac{1}{\omega}$$

such that we can let $\mathfrak{m} = \kappa(a_\omega) = \kappa(b_\omega)$. We are given that f is continuous and so we have continuity at $\mathfrak{m}$ such that $f(a_\omega) \approx f(b_\omega) \approx f(\mathfrak{m})$. Further since $f(x)$ has no zeros (by the supposition) we may manipulate the previous in order to write

$$0 \geq \frac{f(a_\omega)}{f(\mathfrak{m})} \frac{f(\mathfrak{m})}{f(b_\omega)} \approx 1$$

which is a contradiction! Ergo, f must have a zero in $[0, 1]$. ■

¹²depending on whether $f(a) < f(b)$ or $f(a) > f(b)$ for which the sgn adjusts for.

¹³or an odd number of them and thus there exists one

Bibliography

- [1] Michael Barr and Charles Wells, *Toposes, Triples and Theories*, Springer-Verlag, 1985.
- [2] Peter Freyd, *Aspects of Topoi*, BAMS, Volume 7, pp. 1-76, 1972.
- [3] P. T. Johnstone, *Topos Theory*, Academic Press INC. (London) LTD, 1977.
- [4] P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium* Volume 1, Oxford University Press, 2002.
- [5] John L. Kelley, *General Topology*, Springer, 1955.
- [6] Saunders Mac Lane, *Categories for the Working Mathematician*, Springer-Verlag, New York, 1971.
- [7] Saunders Mac Lane and Ieke Moerdijk, *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*, Springer, New York, 1992.
- [8] Abraham Robinson, *Non-Standard Analysis*, North-Holland Publishing Company, Amsterdam, 1966.
- [9] Ross Street, *A Survey of Topos Theory*, 1978.
- [10] K.D. Stroyan and W.A.J. Luxemburg, *Introduction to the Theory of Infinitesimals*, Academic Press INC., New York, 1976.